

WHEN DATA CAN TEACH: PREREQUISITE CIRCUITS GATE LEARNING

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ABSTRACT

The same training data can teach a capability to one model and nothing to another. Curricula and data attribution depend on this effect, yet it remains unclear which parts of a model’s state determine what data can teach. We test whether a previously learned circuit acts as a prerequisite: holding the training data fixed, we suppress an induction circuit only while a transformer trains on a task that builds on it, and we evaluate every model with the circuit intact. In a two-layer transformer, suppressing the circuit’s previous-token head during training blocks learning of long-range copying, and supplying the head’s output restores it; without the circuit, the same data teach nothing for 600,000 updates. In pretrained Pythia models from 160M to 6.9B parameters, suppression blocks or delays learning of a retrieval task that builds on induction, whereas suppressing control heads does not. With longer training, most pretrained models learn the task by rebuilding the retrieval circuit from dormant backup heads. Suppressing these heads after training removes the capability, and suppressing them from the start prevents learning at 410M. The delay, 2 to 18 times the unimpaired learning time, depends on how many backup heads remain. Previously learned circuits can thus determine whether, and how quickly, later data teach a new capability.

1 INTRODUCTION

Whether a batch of training data teaches a model something new depends on the model as well as on the data. Curricula, data selection and data attribution all rely on the fact that the value of an example changes during training (Bengio et al., 2009; Chen et al., 2023; Mindermann et al., 2022; Wang et al., 2025; Lee et al., 2026), yet it is unclear which part of a model’s state decides what data can teach. We ask whether the answer can be a specific learned computation. If a new capability is assembled from computations the model already has, their availability should decide whether a data stream teaches it. We call such computations *prerequisite circuits*.

Testing this idea requires separating two roles a circuit can play. A circuit may be needed to *perform* a capability, which ablation at evaluation time measures, or to *learn* it, which such ablation cannot reveal. We isolate the second role with *training-only suppression*: we silence a circuit only on the training examples of the new task, keep the training batches and their order identical across conditions, and evaluate every model with the circuit intact (Figure 1A). A difference in what the models learn then comes from which computations were available while the data arrived.

Our prerequisite is induction (Olsson et al., 2022), in which a previous-token head and an induction head together retrieve the token that followed an earlier occurrence of the current context. Our target tasks build on it: long-range copying in a two-layer transformer, and a retrieval task with a learned mapping in pretrained Pythia models (Biderman et al., 2023; van der Wal et al., 2025). We make three contributions.

- **A causal test of prerequisite gating** (Section 3). In the two-layer model, suppressing the previous-token head during training blocks learning of long-range copying, and supplying the head’s output restores it. Data that fail to teach succeed when replayed after the circuit forms, and without the circuit the same data teach nothing for 600,000 updates.

(A) Change the computation during training; restore it for evaluation

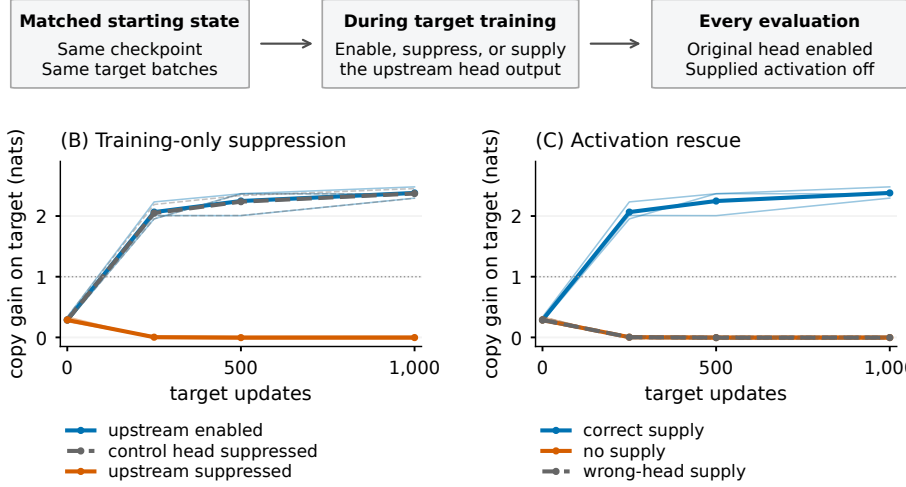


Figure 1: **Suppressing the prerequisite during training blocks learning, and supplying its output restores it** (two-layer model). (A) Every condition starts from the same checkpoint, trains on the same 1,000 target-task batches and is evaluated with the original head enabled and nothing supplied. (B) Suppressing the previous-token head during training blocks learning; suppressing a control head does not. (C) With the head suppressed, supplying its intact output restores learning; supplying nothing or the control head’s output does not. Thin lines: three original models; thick lines: means; dotted lines: the 1-nat criterion. Five further models replicate both results (Appendix P).

- **Gating in a pretrained language model** (Section 4). In Pythia-160M, suppressing four induction heads on the target-task examples blocks learning for 500 updates in all 16 public pretraining runs. Control heads do not, including heads whose removal harms language modeling at least as much.
- **Rebuilding of the prerequisite** (Section 5). From 160M to 6.9B parameters, longer training lets the models learn the target task after 2 to 18 times the control learning time. They rebuild the retrieval circuit from dormant backup heads, which attend to the right positions before training but do not copy. Suppressing these heads removes the capability, the number left unsuppressed sets the delay, and suppressing the rebuilt circuit as it forms keeps key-dependent learning blocked.

Together, the results suggest that a capability built on an existing circuit is learned quickly when that circuit is present, and that without it the data must first rebuild the circuit from whatever parts the model has (Section 6). The two-layer model cannot do so within our budgets; pretrained models can. This links prerequisite learning to self-repair and backup heads (Wang et al., 2023; McGrath et al., 2023; Rushing and Nanda, 2024), to the relearning of removed capabilities (Lo et al., 2024; Jain et al., 2024) and to theories in which learning time depends on steps that must be learned together (Abbe et al., 2021; 2023).

2 SETUP

Prerequisite circuit. In a sequence $\dots AB\dots A$, a first-layer *previous-token head* writes information about A into the position of B . A later *induction head* attends from the second A to that position and copies B (Olsson et al., 2022). The previous-token step forms first (Singh et al., 2024; Bietti et al., 2023; Edelman et al., 2024), and the circuit forms early in pretraining (Olsson et al., 2022; Tigges et al., 2024). The prerequisite is this computation; particular heads implement it. We call a head’s attention from the repeated token to the token that followed its first occurrence its *induction attention*.

Two-layer model. The two-layer model is an attention-only transformer (0.69M parameters, eight heads per layer, context 128) trained on bytes of enwik8, shuffled to keep bigram statistics but destroy

longer-range structure. *Short repeats* (16-byte spans copied 17–31 bytes later) form the induction circuit. The target task is *long repeats* (32-byte spans copied 40–88 bytes later), which reward copying over a greater distance. We measure learning by *copy gain*, the reduction in cross-entropy on copied positions relative to the same positions in unmodified text; a model learns the target task when copy gain reaches 1 nat within 10,000 updates.

Pretrained models. We fine-tune public Pythia checkpoints early in pretraining: step 1000 for 160M to 2.8B, where induction has formed, and step 2000 for 6.9B, whose step-1000 checkpoint falls below our minimum induction accuracy of 0.5. The order experiments start at step 512 of 160M, before induction forms. In the target task, *mapped copy*, a four-token key is followed by one of 32 reserved marker tokens, three distractor markers appear elsewhere, and the key repeats later. The model must output $\pi(m)$, where m is the marker that followed the key and π is a fixed random permutation, so the answer never appears in context. Because choosing the marker nearest the query scores about 60% without reading the key, we test *key dependence* directly. The *key-swap test* exchanges the key with the tokens before a distractor marker, which changes the correct answer, and the *key-deletion test* replaces the key with unused tokens. A model *learns* the target task when accuracy reaches 0.8, it passes the key-swap test on 80% of pairs and deleting the key lowers accuracy by 0.25 or more. The 16-run study in Section 4 uses an accuracy criterion of 0.5 with key tests at the end of training. Each training batch holds six mapped-copy examples, six ordinary induction examples and six windows of natural text.

Interventions. *Training-only suppression* zeros the outputs of selected heads on target-task examples during training; we call the set of suppressed heads the *mask*. In the pretrained models, the other examples in each batch see all heads, all parameters train and every evaluation removes the mask. In the two-layer model, the suppressed head and its input path (embeddings, positions and first normalization) are frozen in every condition, so the restored head computes exactly the same function. *Activation rescue* suppresses the head and adds the output of an intact frozen copy on the same input. *Exact replay* presents the same target-task batches again. *Masked induction* is ordinary induction accuracy measured with the mask applied.

Head selection and controls. We never use target-task outcomes to select heads. In the pretrained models we rank heads by induction attention and suppress the smallest set that lowers held-out induction accuracy to 0.02 or less (0.2 in the 16-run study). *Matched controls* are equally many heads in the same layers with the lowest induction attention. Appendix A gives details, and Appendix B states which choices preceded which outcomes.

3 A PREREQUISITE CIRCUIT GATES LEARNING IN A TWO-LAYER MODEL

The circuit is needed during learning. We start from three models in which the induction circuit has formed and identify it by ablating heads on a short-range copying probe, without reference to the target task. Ablating the selected heads together lowers short-range accuracy to 0.02 or less, whereas ablating equally many control heads leaves 0.72 or more. We then train each model for 1,000 updates on long repeats. With no suppression, or with a control head suppressed, the models learn the target task (copy gain 2.29–2.48 nats). With the previous-token head suppressed during training, copy gain stays between -0.03 and 0.02 nats (Figure 1B). Because the head and its inputs are frozen, the restored head computes the same function at every evaluation, so the conditions differ only in whether the computation was available during training.

The head’s output is what matters. Keeping the head suppressed but adding the output of an intact frozen copy restores learning (copy gain 2.29–2.48 nats), whereas adding nothing or the control head’s output leaves copy gain below 0.023 nats (Figure 1C). The supplied output reproduces the intact forward pass to within 2×10^{-5} in the logits. Five further models, trained from scratch under the same rules, replicate both results (Appendix P).

The same data teach after the circuit forms. One schedule trains on short repeats until the circuit forms and then on 90,000 target-task batches; the other presents the same batches first and the short repeats afterwards. Presented after the circuit forms, the batches yield 2.4 nats or more of copy gain

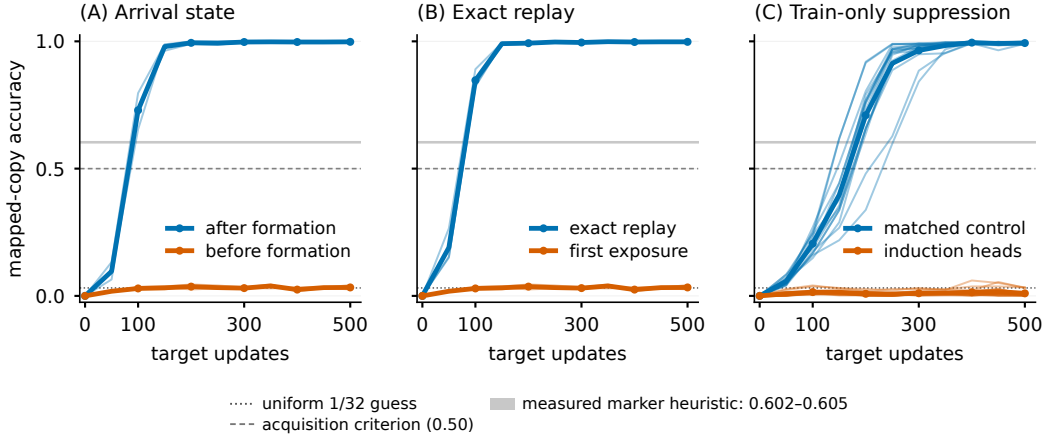


Figure 2: **Order, replay and training-only suppression in Pythia-160M.** Thin lines average the two data streams of one pretraining run; thick lines average runs (three in A and B, all 16 in C). (A) Target-task batches arrive after induction forms (blue) or before (orange). (B) First exposure and exact replay after induction forms. (C) Induction heads (orange) or matched control heads (blue) are suppressed on target-task examples only; every evaluation is unsuppressed. Dotted and dashed lines mark uniform guessing and the 0.5 criterion; the band marks the nearest-marker heuristic.

within 10,000 updates; presented first, they yield at most 0.02 nats after all 90,000. A model of the same age trained only on unmodified text also fails. With exact replay, the first exposure fails in all three seeds. The circuit then forms in two of them, and in both the replayed batches teach the target task by the first evaluation at 500 updates; in the third, the circuit does not form and replay fails (Appendix C, Figure 4). The batches are unchanged, and only the circuitry differs. A four-layer transformer with MLP blocks shows the same order effect (Appendix G).

Without the circuit, the data do not teach. Models trained on the target task first, and models whose previous-token head is reset after the circuit forms, stay below 0.03 nats of copy gain for 600,000 updates, 1,200 times the budget that suffices with the circuit (Appendix E). The two parts of the circuit differ. Resetting the previous-token head (4.8% of parameters) blocks learning for 90,000 updates in six of six models, whereas resetting the selected induction heads lets all six learn within 500 updates; resetting the whole second layer and the readout does not recover within 10,000 updates (Appendix D). In a third cohort, the previous-token reset blocks learning in two of three models. The third keeps a second head with previous-token attention of 0.98 (0.15 or less in every other model), and resetting that head as well blocks learning (Appendix F). What matters is whether some head computes the previous-token step. The target-task data can thus rebuild the induction step but not the previous-token step in this model. In the pretrained models we suppress induction heads on target-task examples, so the induction step cannot be rebuilt in place; Section 5 shows that it is rebuilt in other heads.

4 GATING IN A PRETRAINED LANGUAGE MODEL

Order and replay. At pretraining step 512, induction in Pythia-160M is at chance. We train on natural text with synthetic repeats until induction forms, either before or after 500 target-task batches, in three pretraining runs with two data streams each. When the target task arrives after induction forms, accuracy passes 0.5 within 100 updates and ends at 0.99 or above; presented first, the same stream stays at or below 0.066. After induction then forms, target accuracy is still zero, so induction training alone does not produce the capability. Replaying the same stream then reaches 0.5 within 100 updates in all six cases (Figure 2A–B).

Training-only suppression in 16 pretraining runs. At step 1000 we select four induction heads and suppress them on the mapped-copy examples only. Across the three original runs and the 13

Table 1: **Training-only suppression across model sizes.** *First learned* is the first evaluation (every 50 updates) that meets the learning criterion, for data stream 1 / 2. Masked induction is given at the start and end of training.

Model	Heads suppressed	Updates observed	First learned		Delay	Masked induction start → end
			controls	suppressed		
160M	4	10,000	350	6,400 / 6,300	18×	0.00 → 0.90
410M	8	2,000	150	900 / 900	6×	0.01 → 0.82–0.88
1B	8	2,000	150–200	1,200 / 1,400	6–9×	0.00 → 0.89–0.96
1.4B	7	500	150	partial [†]	—	0.01 → 0.01
2.8B	4	2,000	150–200 [‡]	900 / 900	≈5×	0.01 → 0.92–0.93
6.9B*	24	500	100	200 / 250	2–2.5×	0.00 → 0.88–0.92
410M [§]	8, 4, 4	2,000	150–200	900 / 900; never ^{×4}	6×	0.00 → 0.82, 0.00

*Pretraining step 2000; not comparable to the step-1000 rows. [†]Target accuracy 0.20 and 0.14 at 500 updates, failing the key tests. [‡]Controls from the 500-update study on other data streams. [§]Three independent pretraining runs (run 1; runs 2 and 3); controls from their 500-update study.

other public PolyPythias runs, which vary initialization, data order or both, target accuracy never exceeds 0.067 on either stream, whereas every matched control reaches 0.992 or more within 500 updates (Figure 2C; 32 of 32 paired comparisons; one-sided sign test over runs, $p = 2^{-16}$). The controls’ solutions are key-dependent: they pass the key-swap test on 96–100% of pairs, and deleting the key lowers their accuracy to 2–7% (Appendix H).

Controls that harm language modeling as much still learn. Low-attention control heads also matter little for language modeling, so their failure to block learning might only show that the model tolerates small damage. We therefore select *loss-matched controls*: among heads whose individual removal keeps induction at 90% of its intact accuracy or more, we add heads in order of how much their removal raises natural-text loss, until the set raises it at least as much as the induction heads do. At 160M this takes 16 heads, which raise natural-text loss by 0.43 nats (0.40 for the four induction heads) and leave induction at 0.66 (0.00). Under the protocol of Section 5, suppressing them does not slow learning: the model learns at updates 350 and 400, against 350 without suppression, whereas the four induction heads delay learning to 6,300–6,400 updates. At 410M, eight loss-matched heads give learning at updates 150 and 200, against 150 without suppression (Appendix J).

What suppression removes. Suppressed models also end with lower induction than controls (0.22–0.49 against 0.74 or more), so the requirement concerns the learning trajectory as a whole. Blocking only the target-task gradient through the heads, while keeping their output, does not reliably block learning (Appendix I); the heads’ forward computation is what matters.

5 PRETRAINED MODELS REBUILD THE PREREQUISITE FROM BACKUP HEADS

We repeat training-only suppression at 160M to 6.9B with fresh data streams, longer training and a learning rate of 3×10^{-5} , chosen on control runs only because 10^{-4} destabilized control learning at 410M and 1B. At each size we suppress the smallest set of induction heads, from a coarse grid, that lowers held-out induction to 0.02 or less (4, 8, 8, 7, 4 and 24 heads). Controls learn at every size within 100–350 updates. Because architecture, head width and checkpoint age differ across sizes (Appendix K), we treat each size as a separate case and fit no trend. We did not train 12B, whose step-1000 checkpoint falls below our induction threshold.

Learning is delayed. At 500 updates, the horizon of the 16-run study, suppression blocks learning at 160M, 410M, 1B and 2.8B and in three independent 410M pretraining runs. With longer training, the suppressed models learn the target task at every size except 1.4B, which we train for only 500 updates (Figure 3A; Table 1). 410M and 1B learn after 900–1,400 updates, 6–9 times the control time; 2.8B learns after 900; and 160M, which shows no progress through 3,500 updates, learns after 6,300–6,400, 18 times the control time. Of the three independent 410M runs, one learns at update 900 and two stay blocked for all 2,000 updates.

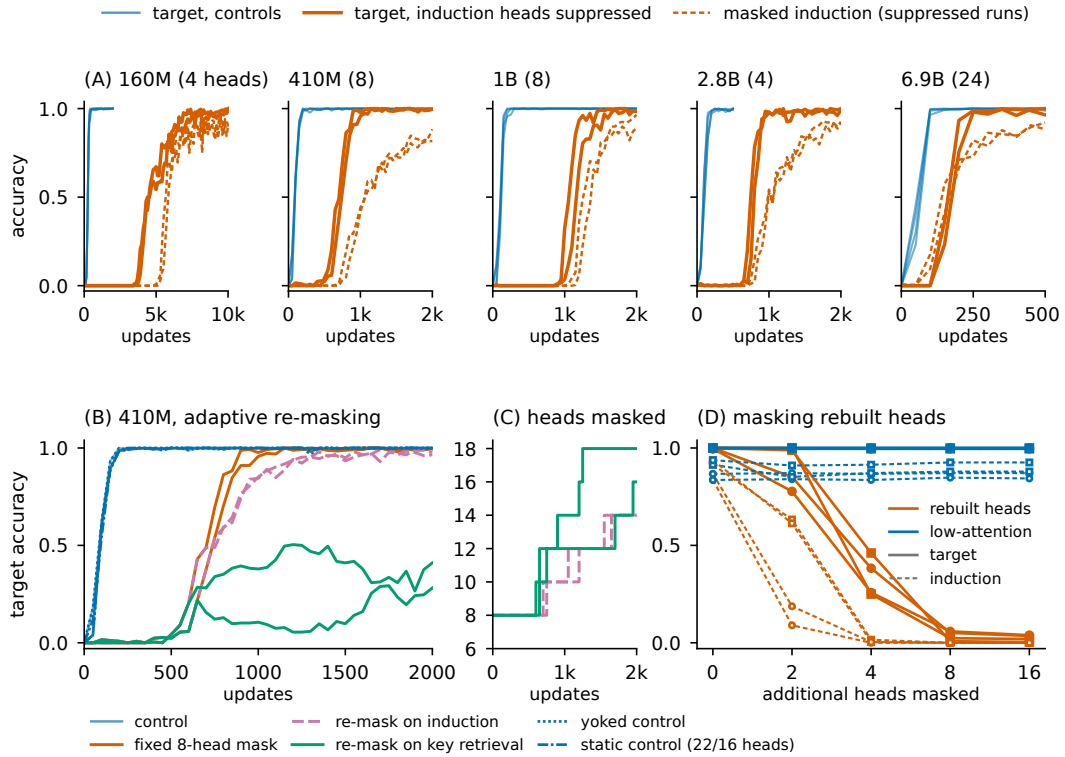


Figure 3: **Pretrained models learn only after rebuilding the prerequisite.** (A) Target accuracy (solid; evaluated without the mask) and masked induction (dashed) for suppressed runs (orange) and controls (blue), two data streams per size; parentheses give the number of suppressed heads. The 2.8B controls come from the 500-update study, and 6.9B starts at pretraining step 2000. (B) Adaptive suppression at 410M. Growing the mask whenever masked induction reappears (pink) does not prevent learning; growing it whenever key-dependent retrieval reappears (green) keeps key-dependent learning blocked. Yoked (dotted) and static (dash-dot) controls with as many low-attention heads suppressed learn by update 150. (C) Mask size in the adaptive runs. (D) After training, suppressing the top rebuilt heads (orange) removes masked induction (dashed) and target accuracy (solid) at 410M (circles) and 1B (squares); suppressing low-attention heads in the same layers (blue) does not.

The rebuilt circuit serves the target task first. Masked induction also recovers, from 0.00 to above 0.8, but after the target task. With the mask applied, target accuracy rises before masked induction at 160M, 410M, 1B and 2.8B: at 410M, masked target accuracy is 0.62 at update 650 while masked induction is 0.004, and at 1B it is 0.59–0.63 while masked induction is zero. Only at 6.9B do the two rise together. Because the mask applies only to target-task examples, the pressure to rebuild comes from the target-task data, which first build a retrieval circuit for the task itself. Four observations show that this circuit is a rebuilt copy of the prerequisite.

(a) The target task depends on the rebuilt heads. After training, we rank the unsuppressed heads of each model that learned by induction attention, with the mask applied, and additionally suppress the top k (Figure 3D). At 410M, suppressing the top four removes held-out induction (0.84–0.87 to 0.00) and lowers target accuracy from 1.00 to 0.26–0.38; suppressing eight lowers it to 0.05–0.06. At 1B, four heads lower target accuracy to 0.25–0.46 and eight to 0.01–0.02. Suppressing equally many low-attention heads in the same layers changes neither. At 6.9B, eleven or twelve additional heads lower induction to 0.01–0.02 and target accuracy to 0.28–0.32 (Appendix K).

(b) The rebuilt heads were dormant backups. In the pretrained 410M and 1B models, these heads rank 9th to 13th by induction attention, just below the suppressed heads. Before training they already attend from the repeated key to its source (induction attention 0.11–0.67 with the mask

applied, against a median of 0.01), yet masked induction is 0.00. We call such heads *backup heads*. Their attention grows during training in every condition, including the controls, but only suppressed training makes their outputs support copying: masked induction reaches 0.84–0.94 after suppressed training and stays at 0.00 after control or unsuppressed training (Appendix M). The prerequisite is thus rebuilt from dormant backup heads (Wang et al., 2023; Gong et al., 2026).

(c) The number of unsuppressed backup heads sets the delay. If learning waits for backup heads to be repurposed, leaving more of them unsuppressed should shorten the delay, and suppressing all of them should prevent learning. Both hold. At 160M, suppressing only the top two induction heads still meets our selection criterion (held-out induction 0.012) but leaves three heads with induction attention above 0.3, and the model learns by update 1,100–1,300. With three heads suppressed it does not learn within 2,000 updates, and with four it needs 6,300–6,400. At 410M, four suppressed heads permit learning within 500 updates and eight delay it to 900; at 1B, the same numbers give learning within 500 updates and at 1,200–1,400. Conversely, suppressing at 410M every head with induction attention of 0.2 or more in the pretrained model (13 heads, fixed before training) keeps target accuracy at or below 0.012 for all 2,000 updates, while unsuppressed induction and natural-text loss stay at the levels of the other runs. Suppressing 13 low-attention heads in the same layers does not slow learning (update 150, as without suppression). The number of backup heads with induction attention of 0.2 or more also separates the three independent 410M runs: run 1 has eight and learns at update 900, while runs 2 and 3 have three each and stay blocked. We counted run 3’s heads before training it. Across sizes, the count orders the delays only roughly (Appendix M).

(d) Suppressing the rebuilt circuit as it forms blocks key-dependent learning. Two adaptive procedures at 410M test whether rebuilding is necessary. Every 50 updates, each procedure measures a detection score with the mask applied and, if the score exceeds a threshold, adds to the mask the unsuppressed heads that attend most from the query to the source until the score falls below it. When the score is masked induction (threshold 0.02), the mask grows to 14 and 16 heads and masked induction never exceeds 0.04, yet the model learns the target task at updates 950 and 1,150 (Figure 3B–C). The rebuilt circuit is specific to the target task, and suppressing general induction does not stop it. When the score is the key-swap pass rate on held-out target-task examples (threshold 0.10), the mask grows to 18 and 16 heads, and key-dependent learning stays blocked for all 2,000 updates (key-swap accuracy at most 0.23 and 0.09). Target accuracy still rises partway, to 0.41 and 0.28, without key dependence. Two controls rule out an effect of suppressing more heads. A yoked control adds equally many low-attention heads in the same layers at the same updates, and a static control suppresses as many low-attention heads as the adaptive runs end with (22 and 16) from the start. Both learn by update 150.

When the circuit cannot be rebuilt in place. The rebuilt circuit must be built somewhere. In a separate 1B experiment we freeze the embeddings and the lower eight of 16 layers, which contain the suppressed heads, and suppress those heads on target-task examples (Appendix N). Supplying their exact outputs during training instead gives the unimpaired learning curve (learning at update 300). With the heads suppressed, learning is delayed nine-fold, to 2,750–2,800 updates, and the model learns a key-dependent workaround in its upper layers that works only while the heads are suppressed: 98% target accuracy with suppression, 2–4% without. Masked induction stays at or below 2/256 throughout, so the workaround does not support ordinary induction on our probe, although it may use computations the probe does not measure.

A stronger intervention, removing attention to the earlier key occurrence in every head, blocks learning for 2,000 updates at 160M to 1.4B but withholds information the task needs, so we report it only in Appendix O.

6 WHY A MISSING PREREQUISITE DELAYS LEARNING

The results suggest one account of both the block in the two-layer model and the delay in pretrained models. A target task that composes an existing computation with something new is learned quickly when the computation is present, because the gradient from target-task examples only has to build the new part. When the computation is missing, the data must first build a replacement, which is a harder learning problem. This resembles theories in which learning time depends on the largest *leap*

of features that must be learned together (Abbe et al., 2021; 2023), and in which intermediate steps make otherwise unlearnable tasks learnable (Wies et al., 2023). The two-layer model localizes the difficulty: it rebuilds a missing induction step within 500 updates but not a missing previous-token step in 600,000, unless a second previous-token head survives.

Pretrained models also need the prerequisite, but they obtain it differently. At 410M and 1B, training on the target task repurposes backup heads that already attend to the right positions. Leaving more of them unsuppressed shortens the delay, and suppressing them all, or suppressing the rebuilt circuit as it forms, blocks key-dependent learning. This agrees with self-repair at inference, where other heads or layers partly compensate for ablated ones (McGrath et al., 2023; Rushing and Nanda, 2024; McDougall et al., 2024), including dormant backups (Wang et al., 2023; Gong et al., 2026), and with the migration of circuit roles between heads during Pythia pretraining (Tigges et al., 2024). It also agrees with relearning after removal: pruned concept neurons are reconstructed elsewhere (Lo et al., 2024), unlearned knowledge returns after brief fine-tuning (Deeb and Roger, 2024; Hu et al., 2025b; Lucki et al., 2025; Jain et al., 2024), and a 160M model whose induction heads are repeatedly reinitialized grows new ones (Sahin et al., 2025). We add that backup capacity decides whether, and how quickly, a new capability that depends on the removed computation can be learned, and that the rebuilt circuit first serves the new task.

Why backup heads are usable remains open. On a lottery-ticket view, training amplifies structure already present (Frankle and Carbin, 2019); here that structure is a pretrained head with the right attention pattern. On an overparameterization view, extra units turn the plateaus of a smaller network into saddles with escape directions (Fukumizu and Amari, 2000; Şimşek et al., 2021). Our data favor the first view in a narrow form, since the rebuilt heads can be identified before training, but do not rule out the second: we do not measure the loss landscape, and model size is confounded with architecture and checkpoint age in Pythia.

Implications. If whether data teach depends on a specific circuit, curricula and data attribution may benefit from conditioning on circuitry (Chen et al., 2023; Lee et al., 2026; Chen et al., 2026). Suppressing a capability’s prerequisite is a weak safeguard in pretrained models, which rebuild it from spare heads; unlearning and safety fine-tuning show a related fragility (Qi et al., 2024; Lee et al., 2024; Che et al., 2025; Tamirisa et al., 2025). Finally, claims that a computation is necessary for learning need a time axis: suppression that blocks learning for 500 updates may only delay it, and a probe of the general prerequisite can miss a task-specific rebuild.

7 RELATED WORK

Internal structure and later learning. Several studies manipulate internal structure during training and observe later capabilities. Chen et al. (2024) suppress syntactic attention in masked language models and find it necessary for later grammatical capabilities; Singh et al. (2024) clamp subcircuits to identify the sub-computations that induction formation requires; Zucchet et al. (2025) supply attention patterns to bypass a plateau in factual learning; and Feng et al. (2025) identify pretrained structures that let later facts generalize. Pretraining data and curricula shape which circuits form and how selectively fine-tuning acts (Bruijns et al., 2026; Chen et al., 2026; Hu et al., 2025a), and fine-tuning often strengthens existing mechanisms (Prakash et al., 2024; Jain et al., 2024). Ablating a few induction heads in pretrained models sharply impairs pattern-matching in-context learning (Crosbie and Shutova, 2025), and Sahin et al. (2025) suppress induction throughout pretraining and find that abstractive in-context learning survives. Our work starts from an already learned circuit, fixes the later data exactly, combines training-only suppression, activation rescue and exact replay under intact evaluation, and follows the dependence across model size and training time.

Training history, data influence and interventions. Influence functions, data valuation and attribution methods quantify how examples affect a trained model (Koh and Liang, 2017; Ghorbani and Zou, 2019; Ilyas et al., 2022; Park et al., 2023), and recent methods track how influence changes during training (Wang et al., 2025; Lee et al., 2026; Bae et al., 2024). Critical periods, warm-starting and plasticity loss show that training history constrains later learning (Achille et al., 2019; Kleinman et al., 2024; Ash and Adams, 2020; Dohare et al., 2024; Lyle et al., 2023), and stagewise development provides internal milestones for such effects (Hoogland et al., 2025; Singh et al., 2023; Reddy, 2024; Chan et al., 2022). Conclusions from ablation and activation patching depend on the method and

metric (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024; Li and Janson, 2024; Miller et al., 2024), and different-looking circuits can implement the same mechanism (Bayat Makou et al., 2026). Gradient routing localizes what is learned with data-dependent gradient masks (Cloud et al., 2024); we instead suppress the forward computation on selected examples while the rest of each batch trains normally.

8 LIMITATIONS AND FUTURE WORK

We study one prerequisite, induction, with two copy-based target tasks; the two-layer model’s data are engineered, and the pretrained-model target task is synthetic. We fine-tune early Pythia checkpoints rather than intervene during pretraining. Architecture, head width, the number of suppressed heads and checkpoint age vary with model size, and above 160M we mostly have one pretraining run per size, so we claim no scaling law. Suppressing a head removes everything it computes, so specificity to induction rests on our controls. Our results depend on the optimizer, learning rate and training horizon, and failing the learning criterion does not imply learning nothing. Appendix Q describes further experiments.

REPRODUCIBILITY STATEMENT

The appendices describe the data construction, model configurations, training protocols, interventions, evaluation criteria, and per-run results. Experiments use deterministic training streams, with batch counts and digests checked for equality across matched conditions. The pretrained-model experiments use public Pythia checkpoints.

AI USE STATEMENT

We used a language-model assistant for literature retrieval and discovery, manuscript drafting and editing, synthetic-data generation code, and assistance with research methodology, implementation, and interpretation of experimental results. The authors take responsibility for the final manuscript and all reported results.

REFERENCES

- Emmanuel Abbe, Enric Boix-Adsera, and Theodor Misiakiewicz. SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics. In *Conference on Learning Theory*, 2023. arXiv:2302.11055.
- Emmanuel Abbe, Enric Boix-Adsera, Matthew S. Brennan, Guy Bresler, and Dheeraj Nagaraj. The staircase property: How hierarchical structure can guide deep learning. In *Advances in Neural Information Processing Systems*, 2021. arXiv:2108.10573.
- Alessandro Achille, Matteo Rovere, and Stefano Soatto. Critical learning periods in deep networks. In *International Conference on Learning Representations*, 2019. arXiv:1711.08856.
- Jordan T. Ash and Ryan P. Adams. On warm-starting neural network training. In *Advances in Neural Information Processing Systems*, 2020. arXiv:1910.08475.
- Juhan Bae, Wu Lin, Jonathan Lorraine, and Roger Grosse. Training data attribution via approximate unrolling. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2405.12186.
- Alireza Bayat Makou, Jingcheng Niu, Subhabrata Dutta, and Iryna Gurevych. Many circuits, one mechanism: Input variation and evaluation granularity in circuit discovery. *Transactions on Machine Learning Research*, 2026. arXiv:2606.06267.
- Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *International Conference on Machine Learning*, pages 41–48, 2009. doi: 10.1145/1553374.1553380.

- Adithya Bhaskar, Alexander Wettig, Dan Friedman, and Danqi Chen. Finding transformer circuits with edge pruning. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.16778.
- Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, 2023. arXiv:2304.01373.
- Alberto Bietti, Vivien Cabannes, Diane Bouchacourt, Hervé Jegou, and Léon Bottou. Birth of a transformer: A memory viewpoint. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.00802.
- Sebastian A. Bruijns, Jirko Rubruck, Mia H. Whitefield, Kai J. Sandbrink, Fazl Barez, and Christopher Summerfield. Pretraining curricula enable selective fine-tuning. *arXiv preprint arXiv:2607.04846*, 2026.
- Stephanie C. Y. Chan, Adam Santoro, Andrew K. Lampinen, Jane X. Wang, Aaditya Singh, Pierre H. Richemond, James L. McClelland, and Felix Hill. Data distributional properties drive emergent in-context learning in transformers. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2205.05055.
- Zora Che, Stephen Casper, Robert Kirk, Anirudh Satheesh, Stewart Slocum, Lev E. McKinney, Rohit Gandikota, Aidan Ewart, Domenic Rosati, Zichu Wu, Zikui Cai, Bilal Chughtai, Yarin Gal, Furong Huang, and Dylan Hadfield-Menell. Model tampering attacks enable more rigorous evaluations of LLM capabilities. *Transactions on Machine Learning Research*, 2025. arXiv:2502.05209.
- Jianhui Chen, Yuzhang Luo, and Liangming Pan. Mechanistic data attribution: Tracing the training origins of interpretable LLM units. In *International Conference on Machine Learning*, 2026. arXiv:2601.21996.
- Mayee F. Chen, Nicholas Roberts, Kush Bhatia, Jue Wang, Ce Zhang, Frederic Sala, and Christopher Ré. Skill-it! A data-driven skills framework for understanding and training language models. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2307.14430.
- Angelica Chen, Ravid Shwartz-Ziv, Kyunghyun Cho, Matthew L. Leavitt, and Naomi Saphra. Sudden drops in the loss: Syntax acquisition, phase transitions, and simplicity bias in MLMs. In *International Conference on Learning Representations*, 2024. arXiv:2309.07311.
- Alex Cloud, Jacob Goldman-Wetzler, Evžen Wybitul, Joseph Miller, and Alexander Matt Turner. Gradient routing: Masking gradients to localize computation in neural networks. *arXiv preprint arXiv:2410.04332*, 2024.
- Joy Crosbie and Ekaterina Shutova. Induction heads as an essential mechanism for pattern matching in in-context learning. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 5049–5111, 2025. arXiv:2407.07011.
- Berfin Şimşek, François Ged, Arthur Jacot, Francesco Spadaro, Clément Hongler, Wulfram Gerstner, and Johanni Brea. Geometry of the loss landscape in overparameterized neural networks: Symmetries and invariances. In *International Conference on Machine Learning*, 2021. arXiv:2105.12221.
- Aggyad Deeb and Fabien Roger. Do unlearning methods remove information from language model weights? *arXiv preprint arXiv:2410.08827*, 2024.
- Shibhansh Dohare, J. Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A. Rupam Mahmood, and Richard S. Sutton. Loss of plasticity in deep continual learning. *Nature*, 632(8026): 768–774, 2024. doi: 10.1038/s41586-024-07711-7.
- Ezra Edelman, Nikolaos Tsilivis, Benjamin L. Edelman, Eran Malach, and Surbhi Goel. The evolution of statistical induction heads: In-context learning Markov chains. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2402.11004.

- Jiahai Feng, Stuart Russell, and Jacob Steinhardt. Extractive structures learned in pretraining enable generalization on finetuned facts. In *International Conference on Machine Learning*, 2025. arXiv:2412.04614.
- Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*, 2019. arXiv:1803.03635.
- Kenji Fukumizu and Shun-ichi Amari. Local minima and plateaus in hierarchical structures of multi-layer perceptrons. *Neural Networks*, 13(3):317–327, 2000. doi: 10.1016/S0893-6080(00)00009-5.
- Amirata Ghorbani and James Zou. Data Shapley: Equitable valuation of data for machine learning. In *International Conference on Machine Learning*, 2019. arXiv:1904.02868.
- Zhiren Gong, He Lu, Tiantong Wang, Yichi Zhang, Yixin Wang, Zihao Zeng, Ming Xiao, Chau Yuen, and Wei Yang Bryan Lim. Conditional co-ablation: Recovering self-repair backups in transformer circuits. *arXiv preprint arXiv:2607.01940*, 2026.
- Michael Hanna, Sandro Pezzelle, and Yonatan Belinkov. Have faith in faithfulness: Going beyond circuit overlap when finding model mechanisms. In *Conference on Language Modeling*, 2024. arXiv:2403.17806.
- Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.
- Jesse Hoogland, George Wang, Matthew Farrugia-Roberts, Liam Carroll, Susan Wei, and Daniel Mufet. Loss landscape degeneracy and stagewise development in transformers. *Transactions on Machine Learning Research*, 2025. arXiv:2402.02364.
- Michael Y. Hu, Jackson Petty, Chuan Shi, William Merrill, and Tal Linzen. Between circuits and Chomsky: Pre-pretraining on formal languages imparts linguistic biases. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9691–9709, 2025a. arXiv:2502.19249.
- Shengyuan Hu, Yiwei Fu, Zhiwei Steven Wu, and Virginia Smith. Unlearning or obfuscating? Jogging the memory of unlearned LLMs via benign relearning. In *International Conference on Learning Representations*, 2025b. arXiv:2406.13356.
- Andrew Ilyas, Sung Min Park, Logan Engstrom, Guillaume Leclerc, and Aleksander Madry. Datamodels: Understanding predictions with data and data with predictions. In *International Conference on Machine Learning*, 2022. arXiv:2202.00622.
- Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P. Dick, Hidenori Tanaka, Edward Grefenstette, Tim Rocktäschel, and David Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. In *International Conference on Learning Representations*, 2024. arXiv:2311.12786.
- Harish Kamath, Emmanuel Ameisen, Isaac Kauvar, Rodrigo Luger, Wes Gurnee, Adam Pearce, Sam Zimmerman, Joshua Batson, Thomas Conerly, Chris Olah, and Jack Lindsey. Tracing attention computation through feature interactions. *Transformer Circuits Thread*, 2025. <https://transformer-circuits.pub/2025/attention-qk/index.html>.
- Michael Kleinman, Alessandro Achille, and Stefano Soatto. Critical learning periods emerge even in deep linear networks. In *International Conference on Learning Representations*, 2024. arXiv:2308.12221.
- Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, 2017. arXiv:1703.04730.
- Andrew Lee, Xiaoyan Bai, Itamar Pres, Martin Wattenberg, Jonathan K. Kummerfeld, and Rada Mihalcea. A mechanistic understanding of alignment algorithms: A case study on DPO and toxicity. In *International Conference on Machine Learning*, 2024. arXiv:2401.01967.

- Jin Hwa Lee, Matthew Smith, Maxwell Adam, and Jesse Hoogland. Influence dynamics and stagewise data attribution. In *International Conference on Learning Representations*, 2026. arXiv:2510.12071.
- Maximilian Li and Lucas Janson. Optimal ablation for interpretability. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2409.09951.
- Michelle Lo, Fazl Barez, and Shay B. Cohen. Large language models relearn removed concepts. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 8306–8323, 2024. arXiv:2401.01814.
- Jakub Łucki, Boyi Wei, Yangsibo Huang, Peter Henderson, Florian Tramèr, and Javier Rando. An adversarial perspective on machine unlearning for AI safety. *Transactions on Machine Learning Research*, 2025. arXiv:2409.18025.
- Clare Lyle, Zeyu Zheng, Evgenii Nikishin, Bernardo Avila Pires, Razvan Pascanu, and Will Dabney. Understanding plasticity in neural networks. In *International Conference on Machine Learning*, 2023. arXiv:2303.01486.
- Flavio Martinelli, Johanni Brea, and Wulfram Gerstner. The puzzling success of overparameterization: Lottery tickets or escape dimensions? Preprint, École Polytechnique Fédérale de Lausanne (EPFL Infoscience), 2026. doi: 10.5075/epfl.20.500.14299/263577.
- Callum Stuart McDougall, Arthur Conmy, Cody Rushing, Thomas McGrath, and Neel Nanda. Copy suppression: Comprehensively understanding a motif in language model attention heads. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 337–363, 2024. arXiv:2310.04625.
- Thomas McGrath, Matthew Rahtz, János Kramár, Vladimir Mikulik, and Shane Legg. The Hydra effect: Emergent self-repair in language model computations. *arXiv preprint arXiv:2307.15771*, 2023.
- Joseph Miller, Bilal Chughtai, and William Saunders. Transformer circuit faithfulness metrics are not robust. In *Conference on Language Modeling*, 2024. arXiv:2407.08734.
- Sören Mindermann, Jan M. Brauner, Muhammed T. Razzak, Mrinank Sharma, Andreas Kirsch, Winnie Xu, Benedikt Höltingen, Aidan N. Gomez, Adrien Morisot, Sebastian Farquhar, and Yarin Gal. Prioritized training on points that are learnable, worth learning, and not yet learnt. In *International Conference on Machine Learning*, 2022. arXiv:2206.07137.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. arXiv:2209.11895.
- Sung Min Park, Kristian Georgiev, Andrew Ilyas, Guillaume Leclerc, and Aleksander Madry. TRAK: Attributing model behavior at scale. In *International Conference on Machine Learning*, 2023. arXiv:2303.14186.
- Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In *International Conference on Learning Representations*, 2024. arXiv:2402.14811.
- Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to! In *International Conference on Learning Representations*, 2024. arXiv:2310.03693.
- Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *International Conference on Learning Representations*, 2024. arXiv:2312.03002.
- Cody Rushing and Neel Nanda. Explorations of self-repair in language models. In *International Conference on Machine Learning*, 2024. arXiv:2402.15390.

- Kerem Sahin, Sheridan Feucht, Adam Belfki, Jannik Brinkmann, Aaron Mueller, David Bau, and Chris Wendler. In-context learning without copying. *arXiv preprint arXiv:2511.05743*, 2025.
- Aaditya K. Singh, Stephanie C. Y. Chan, Ted Moskovitz, Erin Grant, Andrew M. Saxe, and Felix Hill. The transient nature of emergent in-context learning in transformers. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2311.08360.
- Aaditya K. Singh, Ted Moskovitz, Felix Hill, Stephanie C. Y. Chan, and Andrew M. Saxe. What needs to go right for an induction head? A mechanistic study of in-context learning circuits and their formation. In *International Conference on Machine Learning*, 2024. arXiv:2404.07129.
- Rishub Tamirisa, Bhargu Bharathi, Long Phan, Andy Zhou, Alice Gatti, Tarun Suresh, Maxwell Lin, Justin Wang, Rowan Wang, Ron Arel, Andy Zou, Dawn Song, Bo Li, Dan Hendrycks, and Mantas Mazeika. Tamper-resistant safeguards for open-weight LLMs. In *International Conference on Learning Representations*, 2025. arXiv:2408.00761.
- Curt Tigges, Michael Hanna, Qinan Yu, and Stella Biderman. LLM circuit analyses are consistent across training and scale. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2407.10827.
- Oskar van der Wal, Pietro Lesci, Max Müller-Eberstein, Naomi Saphra, Hailey Schoelkopf, Willem Zuidema, and Stella Biderman. PolyPythias: Stability and outliers across fifty language model pre-training runs. In *International Conference on Learning Representations*, 2025. arXiv:2503.09543.
- Jiachen T. Wang, Dawn Song, James Zou, Prateek Mittal, and Ruoxi Jia. Capturing the temporal dependence of training data influence. In *International Conference on Learning Representations*, 2025. arXiv:2412.09538.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023. arXiv:2211.00593.
- Noam Wies, Yoav Levine, and Amnon Shashua. Sub-task decomposition enables learning in sequence to sequence tasks. In *International Conference on Learning Representations*, 2023. arXiv:2204.02892.
- Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, 2024. arXiv:2309.16042.
- Nicolas Zucchet, Jörg Bornschein, Stephanie Chan, Andrew Lampinen, Razvan Pascanu, and Soham De. How do language models learn facts? Dynamics, curricula and hallucinations. In *Conference on Language Modeling*, 2025. arXiv:2503.21676.

A METHODS

A.1 TWO-LAYER MODEL: ARCHITECTURE, DATA AND OPTIMIZATION

The two-layer model is an attention-only causal transformer with width 256, eight heads per layer, context length 128, pre-layer normalization, and 689,664 trainable parameters. Learned position vectors enter the query and key streams but not the value stream. Training minimizes next-byte cross-entropy with batch size 64 and AdamW (learning rate 10^{-3} , betas (0.9, 0.98), weight decay 0.1, 200 warm-up updates). Every condition begins with the same 2,000-update warm-up on unmodified text for its seed.

All corpora derive from one 100 MB order-2 Eulerian shuffle of enwik8, which preserves byte-bigram counts but destroys longer-range structure. The base corpus is this shuffle. The short-repeat corpus copies 16-byte spans forward by a uniformly sampled distance in 17–31 at 400,000 seeded sites. The long-repeat (target) corpus copies 32-byte spans forward by a distance in 40–88 at 170,000 sites. Training crops start below byte 90M; held-out evaluations start at byte 95M. Each corpus has its own deterministic crop-position generator, so moving a block of training in time changes when a

batch is seen but not which batches are seen or in what order. Exact replay rewinds only the target generator. Every result record stores source counts, generator state, corpus hashes and a digest of all crop positions, and the scorers reject any mismatch.

Copy gain (uptake in the released code) is $CE_{\text{base}}(\theta; M) - CE_{\text{target}}(\theta; M)$, where both cross-entropies are computed at the positions M inside pasted spans, on paired crops of the unmodified and target corpora. The key probe presents a four-byte key twice and asks for the byte that followed its first occurrence, at distances 24, 64 and 96 (256 examples each, drawn from the 96 most frequent bytes and generated independently of the training data). Unless stated otherwise, we evaluate every 500 updates, and endpoint values average the final five evaluations.

Formation rule and schedules. After the shared warm-up, the *prerequisite-first* schedule trains on short repeats until two consecutive distance-24 probe evaluations are at least 0.70, no earlier than update 10,000 and with a 120,000-update cap, then trains on 90,000 target batches. The *target-first* schedule consumes the same 90,000 target batches first, then exactly as many short-repeat batches as its paired prerequisite-first run used (Table 2).

Table 2: Order comparison with fixed data in the two-layer model. Copy gain is the maximum in the first 10,000 target updates; the endpoint averages the last five evaluations. *Data match* requires equal target batch counts and position digests within each seed.

seed	formation update	prerequisite first		target first		data match
		copy gain $\leq 10k$	end $d=64$	copy gain $\leq 10k$	end $d=64$	
531	59,500	2.57	0.52	0.00	0.01	yes
532	40,000	2.40	0.52	0.03	0.00	yes
533	84,000	2.58	0.61	-0.03	0.01	yes

A.2 TWO-LAYER MODEL: CIRCUIT IDENTIFICATION AND INTERVENTIONS

In each model in which the prerequisite has formed, we zero each head’s output in turn. The circuit is every head whose individual removal lowers distance-24 accuracy by at least 45% of its intact value, capped at four per layer. Controls are equally many heads with at most a 15% drop, matched by layer where possible (one control in seed 532 comes from the other layer). We set this rule after seeing the first two identification tables and applied it unchanged to the third seed and to all later seeds. Table 3 gives the selected heads and manipulation checks.

Reset redraws a head’s query, key, value, and output weights from the model’s original initialization distribution with a recorded seed; the model then receives a fresh AdamW optimizer and the first 90,000 batches of the same target stream. The conditions are *no edit*; *control reset*; *circuit reset* (all selected heads); *age-matched control* (the warm-up state after an equal number of additional base-corpus updates); *previous-token reset* (the largest-effect first-layer head only); *induction-head reset* (every selected second-layer head); and *heads frozen* (the weights of the six selected heads are fixed, but gradients still pass through them).

Table 3: Per-seed circuit identification. Heads are layer.head. The joint-ablation columns report distance-24 accuracy after zeroing the circuit or the controls.

seed	intact $d=24$	circuit heads	joint ablation	control heads	joint ablation
531	0.73	1.2, 1.7, 1.0, 2.7, 2.3, 2.6	0.012	1.1, 1.4, 1.6, 2.5, 2.2, 2.0	0.72
532	0.72	1.3, 1.2, 1.7, 1.5, 2.6, 2.0	0.004	1.6, 1.4, 1.0, 2.1, 2.3, 2.2	0.81
533	0.73	1.5, 1.6, 1.0, 2.6, 2.5, 2.4	0.020	1.3, 1.7, 1.4, 2.0, 2.2, 2.3	0.82

Training-only suppression with exact restoration. Each model uses its previous-token head (the largest-effect first-layer head) and a low-effect first-layer control head, chosen from the identification records by the smallest absolute change in base-text loss. We freeze the query, key and value rows and output columns of both heads, the token embeddings, the positions and the first normalization. The two heads receive the same first-layer inputs, so this fixes their complete input path (164,352 parameters). Everything else trains.

The unsuppressed, control-suppressed and suppressed conditions receive the first 1,000 target batches. A fourth condition suppresses the previous-token head while training on unmodified text at the same crop positions. Every evaluation enables both heads.

Activation rescue uses the same models and freezing. It suppresses the previous-token head’s own output in every training forward pass and adds one of three signals to the first-layer attention output: the head’s intact contribution, computed by a frozen copy of the starting model on the same tokens; the control head’s contribution; or nothing. Supplied activations are detached. We do not rescale the wrong-head supply, so activation norms are not matched. Every evaluation disables both suppression and supply. The correct supply reproduces the unsuppressed forward pass to within 2.1×10^{-5} in the logits on fixed audit sequences.

A.3 PRETRAINED MODELS

Batches and optimization. One optimizer update uses 18 sequences of length 128, in micro-batches of six, with AdamW (betas (0.9, 0.95), weight decay 0.01, gradient-norm clipping at 1) in FP32. The original 160M studies use learning rate 10^{-4} and 500 updates. The scaling studies use 3×10^{-5} . We chose this rate by a search over $\{3 \times 10^{-5}, 10^{-5}, 3 \times 10^{-6}\}$ that used only unsuppressed and control runs on separate development streams; it is the first rate in the search at which every control learned. In the order and replay study, the induction-forming batch contains six natural-text windows with six copied six-token spans, six random-token sequences with six copied spans and six untouched natural windows (copy distances 24–64).

Target batches and probes. A target batch contains six mapped-copy sequences, six ordinary induction sequences (random tokens with a four-token key repeated 24–64 tokens later) and six natural-text windows. In mapped copy, the token after the first key is one of 32 reserved markers, three distractor markers make the source ambiguous, and the correct continuation of the repeated key is the marker’s image under a fixed random involution. Accuracy is the full-vocabulary argmax at the query position.

Probes contain 256 held-out sequences, and natural-text loss uses 64 held-out windows. Each target-stream seed determines both the data and the involution. The scaling studies use a corrected version of the ordinary-induction training rows, in which the token after the repeated key is the copied continuation; in the original release it was random. The correction changes no qualitative outcome at 160M, 410M or 1B (Appendix K).

Head selection. On an ordinary induction probe (seed 1701), we record each head’s attention from the query (the last token of the repeated key) to the source (the token after the key’s first occurrence). Circuit sets are the top- k heads for increasing k . Controls take, in each layer of the circuit set, the lowest-attention head not already chosen. We select the smallest k for which, on two calibration probes, suppressing the circuit leaves induction accuracy at most 0.02 (0.2 in the original 160M study) with a drop of at least 0.3, while the controls retain at least 0.5 with a drop of at most 0.15. A new 1,024-example probe verifies the choice. Where the lowest-attention controls remove too much induction (1.4B, 6.9B and 12B), we search for layer-matched controls among heads in the lower half of attention scores. Table 22 lists the selections.

Key-dependence tests. The *key-swap* test exchanges the true source’s four-token key with the four tokens before a different-valued distractor in the legal window. This keeps the query, all marker identities and positions, and the multiset of tokens fixed, but changes the correct answer, so no rule that uses only marker identities and positions can answer both versions. Eligible pairs are determined without model outcomes (typically 130–150 of 256 examples). The *key-deletion* test replaces the true key with unused tokens. The nearest-marker heuristic scores 60.2–60.5%.

B WHAT WAS FIXED IN ADVANCE

Head selections use only induction probes, never target outcomes, and we choose learning rates on control and unsuppressed runs only. We wrote down the learning criteria, horizons and evaluation schedules before the corresponding target runs. We report every planned model, seed and pretraining

run, including failures. We calibrated the head-selection threshold of the two-layer model on the first two seeds and applied it unchanged thereafter.

In the two-layer model the replication unit is the initialization and data seed. In Pythia it is the public pretraining run, and the two target streams within a run are technical replicates.

Several analyses are post hoc, and we label them as such where they appear: the timing analysis of delayed learning (Section 5), the per-head attention analyses (Appendix M) and the initial-score comparison in Appendix D. We designed the adaptive-suppression, loss-matched-control, extended-horizon and additional-seed experiments after the main scaling results were known, but fixed their rules before their own outcomes. Interrupted runs resume from saved model, optimizer and data-stream state, and we checked that the overlapping evaluations match exactly.

The follow-up experiments added for this version have the following history. A first key-triggered variant used plain target accuracy under the mask (threshold 0.10) as its detection score. That score sits near its noise floor from the start (0.05–0.10 before any key-dependent learning), so it triggered on noise. We stopped those runs at update 400–500 and replaced the score with the key-swap pass rate before any key-triggered outcome was known; we do not report the stopped runs.

The first 2.8B launch ran out of GPU memory at update 0, and we restarted it from scratch. We stopped the 160M loss-matched runs at update 600 to free memory and reran them from scratch with identical settings; the rerun matches the stopped run at every shared evaluation. We paused one independent 410M run (process suspended) and resumed it without changing its state. We designed the adaptive procedures, the 13-head mask rule, the static and yoked mask controls, the loss-matched selection and the 160M dose runs after seeing that models with fixed masks eventually learn, but fixed their rules before their own outcomes.

C COMPLETE RESULTS FOR THE TWO-LAYER MODEL

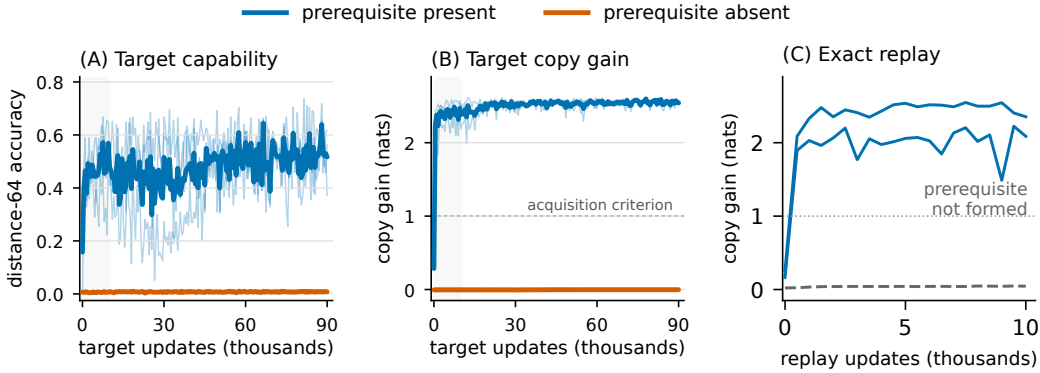


Figure 4: **The same data teach only after the prerequisite forms** (two-layer model). (A–B) Within each seed, both schedules receive the same 90,000 target batches in the same order, either after the induction circuit forms (blue) or before it (orange). Thin lines are three seeds and thick lines are means; the shaded region is the 10,000-update learning window. (C) Exact replay: the model sees the target stream, then short repeats, then the identical target stream again. The two models in which the circuit formed learn on replay; the model in which it did not form (gray) does not.

Training-only suppression and replay afterwards. Table 6 reports every endpoint of the suppression experiment. After the suppressed exposure, the suppressed and unmodified-text conditions receive exact target replay with both heads enabled and fresh optimizers. No model learns immediately when the suppressed head is re-enabled.

Replay afterwards gives mixed results. In seed 531 the model learns after suppressed target exposure but not after unmodified-text exposure, which is compatible with a latent benefit from the suppressed exposure. Seed 532 learns in neither case, and seed 533 learns in both. Re-enabling a head therefore

Table 4: Outcomes of all circuit interventions in the three original models. Start accuracy is measured immediately after the edit. *First* ≥ 1 nat is the first evaluation that crosses the criterion; *none* means no crossing in 90,000 updates.

seed	intervention	start $d=24$	copy gain $\leq 10k$	first ≥ 1 nat	end $d=64$	outcome
531	No edit	0.73	2.56	500	0.47	learned
531	Control reset	0.70	2.53	500	0.61	learned
531	Circuit reset	0.01	0.00	none	0.01	not learned
531	Age-matched control	0.01	0.00	none	0.01	not learned
531	Previous-token reset	0.01	0.00	none	0.01	not learned
531	Induction-head reset	0.02	2.46	500	0.32	learned
531	Heads frozen	0.73	2.50	500	0.60	learned
532	No edit	0.72	2.40	500	0.50	learned
532	Control reset	0.82	2.38	500	0.69	learned
532	Circuit reset	0.01	0.02	none	0.00	not learned
532	Age-matched control	0.01	0.02	none	0.00	not learned
532	Previous-token reset	0.01	0.02	none	0.00	not learned
532	Induction-head reset	0.03	2.39	500	0.59	learned
532	Heads frozen	0.72	2.40	500	0.77	learned
533	No edit	0.73	2.58	500	0.63	learned
533	Control reset	0.76	2.59	500	0.63	learned
533	Circuit reset	0.02	-0.03	none	0.01	not learned
533	Age-matched control	0.01	-0.03	none	0.01	not learned
533	Previous-token reset	0.02	-0.02	none	0.01	not learned
533	Induction-head reset	0.04	2.57	500	0.60	learned
533	Heads frozen	0.73	2.56	500	0.62	learned

Table 5: Exact replay within one model. Copy gain columns are maxima over the full first exposure and the first 10,000 replay updates. *Exact* verifies that the generator was rewound and that both presentations used the same 90,000 batches in the same order.

seed	circuit formed	first exposure	replay $\leq 10k$	first ≥ 1 nat	replay end $d=64$	exact
531	yes	0.00	2.22	500	0.55	yes
532	no (cap)	0.03	0.05	–	0.00	yes
533	yes	-0.02	2.55	500	0.75	yes

guarantees neither that the heads reading its output remain compatible nor that replay recovers within 1,000 updates.

Table 6: Training-only suppression and subsequent replay (copy gain in nats, both heads enabled). Each phase lasts 1,000 updates. The last two columns start from the suppressed-target and unmodified-text endpoints.

seed	unsuppressed	control	suppressed	unmodified text	replay	text + target
531	2.366	2.368	-0.002	-0.005	1.546	0.008
532	2.295	2.292	0.022	0.021	0.032	0.028
533	2.481	2.453	-0.029	-0.034	2.366	2.332

Additional seeds. Five further seeds, trained from scratch under the same rules, replicate both the suppression and the rescue result (Appendix P).

Transfer and restricted plasticity. We transfer the six selected heads, together with the token embeddings, positions, readout and all normalizations, from a model in which the prerequisite has formed into its own earlier (warm-up) checkpoint. The transferred model learns the target task; transferring six control heads with the same interface does not help (Table 8). The transfer changes 52.5% of parameters, and neither the heads alone nor the heads with positions install the prerequisite. The result therefore shows transfer of learned weights, not a minimal circuit transplant.

Table 7: Activation rescue, evaluated with the intact head. Entries are copy gain (nats) / distance-64 accuracy after 1,000 updates.

seed	correct supply	no supply	wrong-head supply
531	2.366 / 0.438	-0.002 / 0.012	-0.002 / 0.012
532	2.295 / 0.336	0.022 / 0.004	0.022 / 0.004
533	2.481 / 0.340	-0.029 / 0.008	-0.030 / 0.008

Holding the weights of the selected heads fixed still permits 1.86–2.36 nats of copy gain by 500 updates, and training only those heads also yields copy gain. The adaptation that follows the circuit therefore has no unique location.

Table 8: Transfer and restricted plasticity (copy gain in nats; the maximum excludes update zero; endpoint accuracy averages the last five evaluations).

seed	intervention	initial	max $\leq 10k$	end $d=64$
531	Circuit transfer	0.142	2.533	0.594
531	Control transfer	-0.001	-0.003	0.014
531	Only heads train	0.259	2.503	0.105
532	Circuit transfer	0.289	2.388	0.634
532	Control transfer	0.017	0.024	0.003
532	Only heads train	0.266	2.417	0.579
533	Circuit transfer	0.390	2.526	0.524
533	Control transfer	-0.042	-0.030	0.007
533	Only heads train	0.337	2.534	0.478

D RESETTING THE PREVIOUS-TOKEN HEAD OR THE INDUCTION HEADS

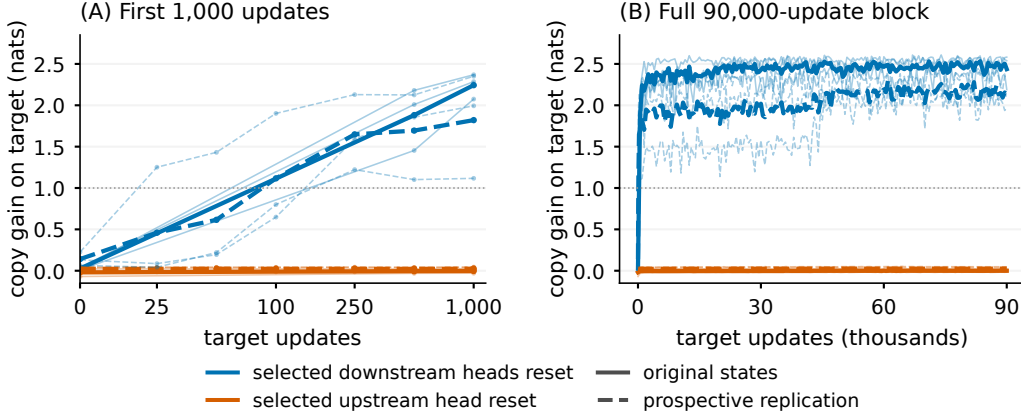


Figure 5: **Resetting the previous-token head blocks learning; resetting the induction heads does not.** All parameters train on the same target stream after either the largest-effect first-layer (previous-token) head is reset (orange) or the selected second-layer (induction) heads are reset (blue). (A) First 1,000 updates (axis linear to 25 updates, logarithmic after). (B) The full 90,000 updates. Thin lines are three original (solid) and three fresh (dashed) models; thick lines are means.

Resetting the previous-token head changes 4.8% of the parameters and changes base-text loss by at most 0.01 nats relative to no edit. In the three original models and three fresh models (seeds 1541–1543, run under the same pre-specified rules), it blocks learning throughout 90,000 target updates. Resetting the selected second-layer (induction) heads instead lets all six models learn within 500 updates (Figure 5). Control resets preserve learning.

Both resets reduce initial short-range accuracy to at most 0.036 in the original models. In the fresh models, however, a pre-specified test of equal initial impairment on the target failed (Table 9): after the induction-head reset the models start slightly better on the target (by 1–8 percentage points of distance-64 accuracy and 0.07–0.23 nats of loss). The comparison therefore shows a difference in recovery between two specific lesions, not between equally impaired models.

A second fresh cohort (seeds 2641–2643) resets *all* eight second-layer heads and the output readout. This broader lesion starts no better than the previous-token reset on either target measure and does not recover within 10,000 updates (Tables 11–12). Damage to the second layer is thus not always easy to repair: within this budget the model rebuilds the selected induction heads but not the whole second layer and readout. In the same cohort the previous-token reset blocks learning in two models and allows it (at update 3,000) in the third, which Appendix F explains.

Table 9: Pre-specified initial matching in the fresh reset models. P and S denote the previous-token reset and the reset of the selected second-layer heads; correct counts are out of 4,096 per condition. Intervals use a simultaneous 0.05 error allocation. Matching requires the accuracy interval within ± 2 percentage points, the loss interval within ± 0.1 nats and both accuracy upper bounds at most 0.05.

seed	correct	P – S interval		floor	
	P/S	accuracy (pp)	target CE (nats)	$\leq .05$	matched
1541	47/163	[-4.21, -1.45]	[0.098, 0.121]	yes	no
1542	31/282	[-7.67, -4.57]	[0.195, 0.233]	no	no
1543	40/99	[-2.59, -0.28]	[0.069, 0.085]	yes	no

Table 10: Learning in the fresh models after the previous-token reset (P) or the reset of the selected second-layer heads (S) over the full 90,000-update stream, and with no edit or a control reset (I/C) over 10,000. *S first* is the first observed copy gain ≥ 1 nat; no P condition reaches it.

seed	P max	S at 500	S first	end $d=64$ P/S	end copy gain I/C
1541	0.043	1.856	250	0.008/0.605	2.154/2.158
1542	0.054	2.125	25	0.012/0.516	2.054/2.085
1543	-0.008	1.101	250	0.004/0.613	1.936/1.931

Table 11: Initial impairment in the broader-lesion cohort (P is the previous-token reset; S resets all second-layer heads and the readout). Gaps are S minus P with paired 95% intervals.

seed	accuracy P/S	loss P/S	accuracy gap (pp)	loss gap (nats)
2641	0.008/0.005	2.703/5.799	[-0.659, 0.024]	[3.081, 3.112]
2642	0.010/0.004	2.704/5.726	[-0.965, -0.256]	[3.007, 3.038]
2643	0.011/0.004	2.713/5.705	[-1.093, -0.323]	[2.976, 3.007]

Table 12: Broader-lesion cohort over 10,000 target updates.

seed	intact first	first P/S	end copy gain P/S	end $d=64$ P/S
2641	25	none/none	0.010/0.008	0.012/0.012
2642	50	3000/none	1.472/0.015	0.363/0.023
2643	250	none/none	-0.020/-0.020	0.008/0.008

E LONG HORIZON

For each original seed, the target-first model repeats the warm-up and trains continuously from target batch zero to 600,000. The previous-token-reset model restores its archived 90,000-update model and optimizer state, verifies the data-stream position and trains a further 510,000 updates. We evaluate every 500 updates. A memory fault at launch required restarting from the warm-up and the archived

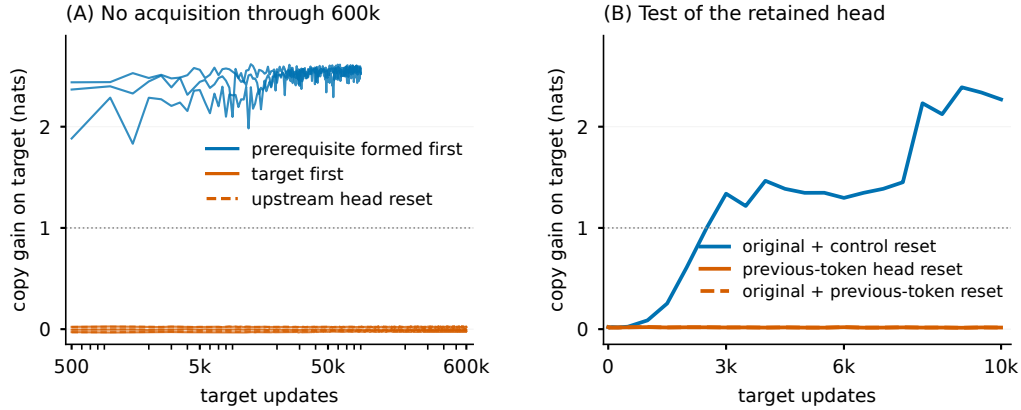


Figure 6: **Persistent failure and the retained previous-token head.** (A) Six continuations without the prerequisite (target first and previous-token reset, three seeds each) stay below 0.03 nats through 600,000 updates; with the prerequisite, the model learns within 500 updates (logarithmic axis). (B) In the one model that learned after the previous-token reset, resetting its remaining previous-token head blocks learning; an equally large control reset does not.

endpoint, respectively; no outcome informed the restart, and we replace no seed (Table 13). The 1,200-fold comparison uses learning by 500 updates when the prerequisite is present.

Table 13: Long-horizon continuations without the prerequisite. End values average the last five evaluations.

seed	condition	updates	max copy gain	first ≥ 1 nat	end copy gain	end $d=64$
531	target first	600,000	0.00	none	-0.01	0.01
531	previous-token reset	600,000	0.00	none	0.00	0.01
532	target first	600,000	0.03	none	0.02	0.00
532	previous-token reset	600,000	0.03	none	0.02	0.00
533	target first	600,000	-0.02	none	-0.02	0.01
533	previous-token reset	600,000	-0.01	none	-0.02	0.01

F THE RETAINED PREVIOUS-TOKEN HEAD

For each of the nine models with a previous-token reset, Table 14 gives the largest mean previous-token attention of any first-layer head immediately after the reset. We made this measurement after all outcomes were known, so it is descriptive. In the one model that learned (2642), the head that the selection rule reset had tied with another head on ablation effect, and the other head has previous-token attention of 0.981.

We then returned to that model’s checkpoint after the circuit formed and, under a protocol written in advance, reset (i) the retained previous-token head, (ii) both it and the originally reset head, or (iii) the original head and a low-effect control head (equal parameter count to ii). Only (iii) learns (Table 15; Figure 6B). A thermal abort interrupted the first attempt; the complete rerun used the same checkpoint, protocol and criteria.

G FOUR-LAYER MODEL

We repeat the order comparison in two seeds of a four-layer pre-normalized transformer with MLP blocks (width 256, eight heads, MLP width 1,024), with the same formation rule and 60,000-update target blocks. Presenting the target after the circuit forms yields 2.58/2.55 nats of copy gain and 0.55/0.66 distance-64 accuracy at the end; presenting it first yields 0.001/0.005 nats and 0.010/0.015.

Table 14: All nine models with a previous-token reset: largest previous-token attention remaining in any first-layer head after the reset.

cohort	seed	remaining prev.-token attention	horizon	first ≥ 1 nat
original	531	0.148	90,000	none
original	532	0.015	90,000	none
original	533	0.102	90,000	none
fresh	1541	0.011	90,000	none
fresh	1542	0.011	90,000	none
fresh	1543	0.011	90,000	none
broader lesion	2641	0.011	10,000	none
broader lesion	2642	0.981	10,000	3,000
broader lesion	2643	0.011	10,000	none

Table 15: Targeted resets in model 2642 (10,000 updates).

reset	remaining attention	first ≥ 1 nat	end copy gain	end $d=64$
previous-token head	0.015	none	0.016	0.023
original + previous-token	0.011	none	0.016	0.020
original + control	0.981	3,000	2.269	0.617

H PYTHIA-160M: COMPLETE RESULTS

Order and exact replay. The order study starts from Pythia-160M at step 512. Induction training ends after two consecutive ordinary-induction accuracies of at least 0.70 (minimum 300 updates, cap 6,000). Target blocks last 500 updates, and we evaluate every 50 updates (Table 16).

Table 16: Pythia-160M order and replay. k is the induction-training length; *pre-replay* is target accuracy after induction training but before the target stream is shown again. Streams 8 and 9 are paired within each run.

run	stream	k	target after induction		target first		exact replay		result
			first ≥ 0.5	end	max	pre-replay	first ≥ 0.5	end	
base	8	2,200	100	1.000	0.066	0.000	100	0.996	pass
base	9	2,200	100	1.000	0.039	0.000	100	1.000	pass
seed 1	8	2,400	100	0.992	0.062	0.000	100	1.000	pass
seed 1	9	2,400	100	0.996	0.047	0.000	100	0.996	pass
seed 2	8	2,500	100	1.000	0.043	0.000	100	1.000	pass
seed 2	9	2,600	100	1.000	0.051	0.000	100	0.996	pass

Training-only suppression. At step 1000 of the same three runs, the selection rule (0.2 ceiling) chooses four heads. Before training, suppressing them lowers induction from at least 0.699 to at most 0.023, while the controls retain at least 0.699. Each condition starts from the public checkpoint with a fresh optimizer and consumes 500 batches of the same stream. A pre-hook zeros the selected head channels before the attention output projection, only for the six mapped-copy examples in each batch.

Target accuracy under suppression never exceeds 0.047, while every control reaches at least 0.996 (Table 17). The suppressed models end with lower (unsuppressed) ordinary induction than the controls (0.22–0.49 against at least 0.74), while their natural-text loss differs by at most 0.024 nats. The intervention therefore changes how induction itself develops, and we interpret it as a requirement on the learning trajectory. Evaluated *with* the suppression applied, the suppressed models also stay below 0.10 (Table 19).

Thirteen further pretraining runs. We apply the identical protocol to the thirteen remaining public PolyPythias 160M runs at step 1000: seven with new initialization and data order (seeds 3–9), three with the original weights and a new data order, and three with the original data order and new

Table 17: Pythia-160M training-only suppression (original three runs). *Induction probe* is the manipulation check before training. Target columns are maxima over 500 updates, evaluated without suppression.

run	stream	induction probe			target maximum		control first ≥ 0.5	result
		intact	suppressed	control	suppressed	control		
base	8	0.699	0.000	0.699	0.012	1.000	200	pass
base	9	0.699	0.000	0.699	0.027	0.996	200	pass
seed 1	8	0.734	0.023	0.730	0.016	1.000	200	pass
seed 1	9	0.734	0.023	0.730	0.047	1.000	200	pass
seed 2	8	0.730	0.004	0.738	0.020	1.000	200	pass
seed 2	9	0.730	0.004	0.738	0.020	0.996	200	pass

weights. Eligibility requires induction of at least 0.5, suppressed induction of at most 0.2 with a drop of at least 0.3, control induction of at least 0.5 with a drop of at most 0.15, and initial target accuracy of at most 0.1. All thirteen runs are eligible, and all pass on both streams (Table 18).

Table 18: The thirteen additional pretraining runs (streams 8/9). Target columns are maxima over 500 updates. The last two columns evaluate the controls at the end; *key swap* requires both answers of a pair to be correct.

run	induction	target: suppressed	target: control	key swap	key deleted
seed 3	0.691	0.055/0.055	1.000/1.000	0.993/0.993	0.051/0.039
seed 4	0.715	0.027/0.023	1.000/1.000	0.986/0.993	0.039/0.031
seed 5	0.676	0.031/0.047	0.996/1.000	0.986/0.993	0.043/0.047
seed 6	0.672	0.051/0.066	0.996/1.000	0.993/0.986	0.023/0.043
seed 7	0.711	0.023/0.027	1.000/0.992	0.993/0.993	0.039/0.023
seed 8	0.641	0.062/0.059	0.996/1.000	0.959/0.986	0.062/0.047
seed 9	0.723	0.016/0.020	1.000/1.000	0.966/1.000	0.027/0.070
data 1	0.680	0.035/0.020	0.996/0.996	0.986/0.986	0.047/0.020
data 2	0.664	0.008/0.016	1.000/1.000	0.986/1.000	0.051/0.035
data 3	0.734	0.016/0.020	0.996/0.996	0.980/0.972	0.074/0.043
weight 1	0.738	0.020/0.031	1.000/1.000	0.980/0.993	0.074/0.047
weight 2	0.711	0.020/0.020	0.996/1.000	0.966/1.000	0.055/0.047
weight 3	0.680	0.027/0.031	0.996/1.000	0.993/0.993	0.043/0.051

Table 19: Key dependence and evaluation mode in the six original run-stream pairs. Suppressed columns are maxima over updates 0–500 with the suppression off or on at evaluation; control columns are at update 500. Pair counts are 147 (stream 8) and 142 (stream 9).

run	stream	suppressed, maximum		control, endpoint		
		off	on	intact	key deleted	key swap
base	8	0.012	0.074	0.984	0.047	0.980
base	9	0.027	0.051	0.996	0.035	0.993
seed 1	8	0.016	0.098	0.992	0.043	0.993
seed 1	9	0.047	0.094	0.996	0.043	0.986
seed 2	8	0.020	0.062	0.996	0.055	0.986
seed 2	9	0.020	0.051	0.988	0.039	0.965

I BLOCKING ONLY THE GRADIENT THROUGH THE PREREQUISITE

To separate the forward computation of the induction heads from their plasticity, we keep the heads’ forward outputs on target examples but detach them (and their output-projection columns) from the

target loss, so that target examples cannot send gradient through these paths. Ordinary induction and natural-text examples update all parameters normally. Four of the six run-stream pairs exceed 0.5 accuracy, but only one passes both key criteria (Table 20). Some models therefore learn substantially without target gradient through these heads, and the suppression result in Section 4 depends on removing the forward computation.

Table 20: Target-gradient blocking with the induction heads’ forward computation kept. All accuracies are measured without the intervention.

run	stream	first ≥ 0.5	end target	key swap	key deleted
base	8	500	0.621	0.381	0.039
base	9	500	0.613	0.423	0.047
seed 1	8	450	0.680	0.483	0.020
seed 1	9	450	0.895	0.831	0.031
seed 2	8	none	0.039	0.000	0.051
seed 2	9	none	0.016	0.000	0.031

J LOSS-MATCHED CONTROLS

Selection uses only the calibration probes (seed 1701): 256 ordinary induction sequences and 64 natural-text windows. For every head we measure natural-text loss and induction accuracy with that head alone zeroed. A head is eligible if its individual removal keeps induction at 90% or more of its intact value. We add eligible heads in decreasing order of their individual loss increase until the set contains at least as many heads as the circuit and its joint removal increases natural-text loss at least as much as the circuit’s joint removal.

Table 21 reports the selections and training outcomes under the scaling-study protocol (3×10^{-5} , 2,000 updates, streams 38/39). Individually, induction heads cost little natural-text loss (0.006–0.046 nats each at 160M), but jointly they cost 0.40 nats, which is why many non-induction heads are needed to match them.

Table 21: Loss-matched controls. Loss increase and induction are measured with the set jointly zeroed on the calibration probes. *First learned* uses the full key-dependent criterion (streams 38/39). *From the 10,000-update runs; the end target for that row is at 10,000 updates.

model	set	heads	loss increase (nats)	induction	first learned	end target
160M	none	0	—	0.70	350 / 350	1.00 / 1.00
160M	induction	4	0.40	0.00	6,400 / 6,300*	0.99 / 1.00
160M	loss-matched	16	0.43	0.66	350 / 400	1.00 / 1.00
410M	none	0	—	0.61	150 / 150	1.00 / 1.00
410M	induction	8	0.23	0.01	900 / 900	1.00 / 1.00
410M	loss-matched	8	0.27	0.61	150 / 200	1.00 / 1.00

K SCALING STUDY: DETAILS AND COMPLETE RESULTS

Eligibility. At step 1000, 6.9B and 12B do not meet the pre-specified requirement that unsuppressed induction be at least 0.5 on both calibration probes (0.37/0.39 and 0.50/0.48), so we do not train them at step 1000. A separately specified search selected step 2000 for both. We trained 6.9B; we did not train 12B, for lack of storage. The number of heads needed for near-complete suppression is not monotonic in size, and head counts are not comparable across architectures, because head width and count differ.

Head count. The delay depends on how completely induction is suppressed at the start. On fresh streams 38/39 at 410M with the original training labels, suppressing four heads (leaving 6–8% induction) permits key-dependent learning by update 400/450 (end accuracy 0.95/0.98), while eight heads (0.6%) block it at 500 updates (0.004/0.004). With corrected labels the four-head runs also learn within 500 updates (0.96/0.98), and the eight-head runs learn at update 900.

Table 22: Models and selected interventions. Holdout values are ordinary induction accuracy on a 1,024-example probe used only after selection (intact / circuit suppressed / control suppressed).

model	step	layers $\times$ heads	head dim	suppressed	holdout
160M	1000	12×12	64	4	0.772 / 0.004 / 0.786
410M	1000	24×16	64	8	0.627 / 0.006 / 0.664
1B	1000	16×8	256	8	0.803 / 0.003 / 0.712
1.4B	1000	24×16	128	7	0.821 / 0.017 / 0.833
2.8B	1000	32×32	80	4	0.711 / 0.004 / 0.715
6.9B	2000	32×32	128	24	0.700 / 0.017 / 0.594
12B	2000	36×40	128	32	0.793 / 0.006 / 0.793

At 1.4B, four heads permit learning by update 350 (end accuracy 0.95/0.99), and seven give partial learning that fails the key test (0.20/0.14 accuracy, key swap 0.02/0.01). At 1B on streams 78/79, four heads permit learning (0.99/0.93) with masked induction recovering to 0.73–0.82, and eight heads block it (0.00/0.00). In every case the controls learn.

Independent 410M pretraining runs. In three further public 410M runs (step 1000; 8, 4 and 4 heads selected), the suppressed models stay at 0.00–0.02 target accuracy through 500 updates on both streams, with masked induction at or below 0.004, and all twelve controls learn. Over 2,000 updates (Table 23), run 1 (eight heads suppressed) learns at update 900 on both streams, like the reference run. Runs 2 and 3 (four heads each) stay blocked on both streams: target accuracy never exceeds 0.10 and 0.004, and masked induction never exceeds 0.004.

The three runs leave 8, 3 and 3 unsuppressed heads with induction attention of at least 0.2 (strongest 0.45, 0.48 and 0.28). We computed these counts while run 2 was in progress and before run 3 began, and predicted that run 3 would learn later than run 1.

Table 23: Independent 410M pretraining runs over 2,000 updates (stream 38 / 39). *Masked induction* > 0.2 is the first evaluation at which ordinary induction with the suppression mask applied exceeds 0.2.

run	heads	first learned	masked induction > 0.2	end target	end masked induction
run 1	8	900 / 900	800 / 850	0.99 / 1.00	0.84 / 0.80
run 2	4	never / never	never / never	0.04 / 0.10	0.00 / 0.00
run 3	4	never / never	never / never	0.00 / 0.00	0.00 / 0.00

Corrected induction examples. Correcting the ordinary induction training rows (Appendix A.3) preserves every qualitative outcome at 160M (4 heads), 410M (4 and 8 heads) and 1B (8 heads) on both streams: the suppressed models’ target accuracy is 0/0, 0.96/0.98, 0.04/0.02 and 0/0, respectively, and the controls learn.

2,000-update horizon. Table 24 gives endpoints of the 2,000-update runs. The first 500 updates of each run reproduce the earlier 500-update runs exactly.

Table 24: 2,000-update endpoints (streams 38/39 for 160M and 410M; 78/79 for 1B). *Masked induction* applies the original suppression mask at evaluation.

model	condition	target	key swap	key deleted	masked induction
160M	suppressed	0.000 / 0.000	0.000 / 0.000	0.023 / 0.023	0.000 / 0.000
160M	control	1.000 / 1.000	1.000 / 1.000	0.047 / 0.031	—
410M	suppressed	0.996 / 0.996	0.993 / 0.993	0.004 / 0.004	0.883 / 0.816
410M	control	0.996 / 1.000	1.000 / 1.000	0.020 / 0.008	—
1B	suppressed	0.988 / 0.984	0.984 / 0.987	0.000 / 0.000	0.891 / 0.965
1B	control	1.000 / 1.000	1.000 / 1.000	0.000 / 0.000	—

Longer horizon at 160M and 2,000 updates at 2.8B. We continue the 160M suppressed runs to 10,000 updates (evaluated every 100) and the 2.8B suppressed runs to 2,000 updates, on the same corrected streams 38/39, whose first 2,000 batches are byte-identical to those of the 2,000-update runs. Table 25 gives their outcomes. Both 160M runs first learn at 6,400 and 6,300 updates. In both, target accuracy passes 0.1 at updates 3,800–4,000, while masked induction first exceeds 0.02 only at 5,100–5,300. We did not rerun the 2.8B controls on these streams; controls on the 500-update streams learned by update 150–200.

Where the rebuilt circuit lives at 6.9B. With the original 24 heads suppressed, we rank the remaining heads of each 6.9B model that learned by induction attention on fresh probes and add heads until induction falls to at most 0.02. Twelve (stream 58) or eleven (stream 59) additional heads suffice, verified on a 1,024-example holdout (induction to 0.01–0.02). Adding them lowers target accuracy from 1.00 to 0.28/0.32 and key-swap accuracy to 0.13/0.09, while layer-matched controls leave target accuracy at 1.00/0.97. The additional heads cost 0.09/0.05 nats of natural-text loss relative to the controls. In the pretrained 6.9B model the same 36 heads remove induction entirely, so this analysis cannot tell whether they were capable of induction beforehand.

Distance coverage. Induction probes at distances 24, 44, 64 and 96, in native and position-matched versions, show the same pattern. The trained 160M suppressed models score zero with the mask applied at every distance. The 410M and 1B suppressed models recover masked induction at all distances, including 96, which lies outside the training range (recovery there is weaker). The fixed 24-head mask at 6.9B leaves under 1.8% induction initially across all eight probes, and 66–92% after training.

L ADAPTIVE SUPPRESSION

Both adaptive procedures start from the fixed 8-head 410M intervention and use the 2,000-update streams 38/39. Every 50 updates, after evaluation, the procedure measures a detection score with the current mask applied. If the score exceeds the threshold, the procedure repeatedly adds the two unsuppressed heads with the highest query-to-source attention (under the current mask) until the score is back at or below the threshold.

Induction-triggered suppression uses ordinary induction accuracy on the two calibration probes as the score (threshold 0.02, the selection criterion). *Key-triggered* suppression uses the key-swap pass rate on two held-out mapped-copy probes, generated with the stream’s mapping but disjoint from the evaluation probes (threshold 0.10). This score responds only to key-dependent retrieval of the target, not to shortcuts.

A *yoked control* replays the key-triggered schedule of additions (same updates, same layers, same counts) but adds the lowest-attention heads in those layers. Because the yoked controls learn before the first addition, we also run a *static control* that masks, from update 0, as many layer-matched lowest-attention heads (attention at most 0.013) as the key-triggered runs end with (22 and 16). Until the first addition, the adaptive runs reproduce the fixed-mask runs exactly (checked at every evaluation).

Table 25 reports these runs together with the other follow-up experiments.

M WHERE THE REBUILT CIRCUIT LIVES AT 410M AND 1B

For each 2,000-update endpoint we measure, with the original mask applied, every head’s query-to-source attention on the two calibration probes. We then mask the top- k remaining heads in addition to the original set ($k \in \{2, 4, 8, 16\}$), or k layer-matched low-attention heads, and measure induction on a held-out probe (seed 4701, never used for ranking), target accuracy and the key tests.

Table 26 lists, for each endpoint, the four unsuppressed heads with the highest attention after training, their rank by attention in the pretrained model (unsuppressed, calibration probes), and their attention before and after training. The same heads top the list in every condition, and they are ranks 9–13 in the pretrained model, immediately below the suppressed set. Their attention grows in every condition,

Table 25: All follow-up runs (streams 38/39; learning rate 3×10^{-5} ; values are stream 38 / stream 39). *Heads* is the number of suppressed heads at the end. *First learned* uses the full key-dependent criterion, evaluated every 50 updates (every 100 for the 10,000-update runs). *Masked induction* is ordinary induction at the end with the run’s final mask applied.

condition	heads	updates	first learned	end target	end key swap	masked induction
160M, 4 heads, 10,000 updates	4 / 4	10,000 / 10,000	6,400 / 6,300	0.99 / 1.00	1.00 / 0.99	0.90 / 0.90
160M, top 2 heads	2 / 2	2,000 / 2,000	1,300 / 1,100	0.96 / 0.99	0.92 / 1.00	0.89 / 0.79
160M, top 3 heads	3 / 3	2,000 / 2,000	never / never	0.00 / 0.03	0.00 / 0.00	0.00 / 0.00
160M, 16 loss-matched heads	16 / 16	2,000 / 2,000	350 / 400	1.00 / 1.00	1.00 / 1.00	0.97 / 0.93
410M, 13 heads (attention ≥ 0.2)	13 / 13	2,000 / 2,000	never / never	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
410M, 13 matched control heads	13 / 13	2,000 / 2,000	150 / 150	1.00 / 1.00	1.00 / 1.00	0.95 / 0.93
410M, loss-matched heads	8 / 8	2,000 / 2,000	150 / 200	1.00 / 1.00	1.00 / 1.00	0.94 / 0.95
410M, adaptive (induction trigger)	14 / 16	2,000 / 2,000	950 / 1,150	0.98 / 0.96	0.97 / 0.95	0.00 / 0.03
410M, adaptive (key trigger)	22 / 16	2,000 / 2,000	never / never	0.41 / 0.28	0.13 / 0.09	0.00 / 0.00
410M, yoked control	22 / 16	2,000 / 2,000	150 / 150	1.00 / 1.00	1.00 / 1.00	0.93 / 0.96
410M, static control (22 / 16 heads)	22 / 16	2,000 / 2,000	150 / 150	1.00 / 1.00	1.00 / 1.00	0.96 / 0.95
2.8B, 4 heads, 2,000 updates	4 / 4	2,000 / 2,000	900 / 900	0.99 / 1.00	0.99 / 1.00	0.93 / 0.92

but only suppressed training makes them produce induction when the original heads are masked (last column). Table 27 gives the ablation results.

We designed this analysis after seeing that the suppressed models learn. The ranking uses the endpoint attention, and the rank comparison is descriptive. The prospective test in Section 5 fixes the mask from pretrained attention alone.

Table 26: The heads that carry induction after training, with the original mask applied. *Rank* is the head’s rank by induction attention in the pretrained model (the suppressed heads are ranks 1–8). Attention before and after training is measured with the original heads masked. *Masked induction* is held-out induction accuracy with the original suppressed heads zeroed.

model	stream	training	top heads after training	rank	attention before	attention after	masked induction
410M	38	circuit	11.1, 10.0, 7.7, 8.3	9, 11, 10, 12	0.67, 0.58, 0.53, 0.34	0.95, 0.90, 0.88, 0.85	0.87
410M	38	control	10.0, 11.1, 7.7, 8.3	11, 9, 10, 12	0.58, 0.67, 0.53, 0.34	0.92, 0.89, 0.83, 0.74	0.00
410M	38	intact	11.1, 10.0, 7.7, 8.3	9, 11, 10, 12	0.67, 0.58, 0.53, 0.34	0.92, 0.88, 0.87, 0.69	0.00
410M	39	circuit	11.1, 7.7, 10.0, 9.6	9, 10, 11, 13	0.67, 0.53, 0.58, 0.27	0.92, 0.89, 0.89, 0.88	0.84
410M	39	control	10.0, 11.1, 7.7, 8.3	11, 9, 10, 12	0.58, 0.67, 0.53, 0.34	0.88, 0.86, 0.84, 0.67	0.00
410M	39	intact	10.0, 11.1, 7.7, 8.3	11, 9, 10, 12	0.58, 0.67, 0.53, 0.34	0.93, 0.93, 0.88, 0.81	0.00
1B	78	circuit	6.4, 7.2, 6.6, 8.5	12, 13, 9, 11	0.16, 0.17, 0.32, 0.11	0.91, 0.87, 0.86, 0.85	0.91
1B	78	control	6.6, 6.4, 5.2, 7.2	9, 12, 10, 13	0.32, 0.16, 0.20, 0.17	0.79, 0.76, 0.72, 0.46	0.00
1B	78	intact	6.6, 6.4, 5.2, 7.2	9, 12, 10, 13	0.32, 0.16, 0.20, 0.17	0.73, 0.70, 0.67, 0.41	0.00
1B	79	circuit	6.6, 7.2, 6.4, 5.2	9, 13, 12, 10	0.32, 0.17, 0.16, 0.20	0.94, 0.92, 0.91, 0.91	0.94
1B	79	control	6.6, 5.2, 6.4, 7.2	9, 10, 12, 13	0.32, 0.20, 0.16, 0.17	0.79, 0.71, 0.66, 0.46	0.00
1B	79	intact	6.6, 6.4, 5.2, 7.2	9, 12, 10, 13	0.32, 0.16, 0.20, 0.17	0.79, 0.66, 0.63, 0.48	0.00

Latent induction heads across models. Table 28 counts, in each pretrained model, the heads outside the selected suppression set that attend from the repeated key to its source. Every model has some. Ordered by the count of heads at or above 0.2 attention, the suppressed models learn roughly as follows. With one such head (160M), learning takes 6,300–6,400 updates; with two (1B), 1,200–1,400; with three (160M with only two heads suppressed), 1,100–1,300, whereas independent 410M runs 2 and 3, also with three, do not learn within 2,000; and with five to eight (410M reference and run 1, 2.8B), about 900.

We computed the count for 160M, 410M, 1B, 1.4B and 2.8B after observing that the suppressed models learn (except at 2.8B, whose long runs came later), and for the independent 410M runs while run 2 was in progress and before run 3 began. The relation is only roughly monotone and is confounded by architecture, head width and checkpoint, so we treat it as suggestive.

N FREEZING THE LOWER HALF OF PYTHIA-1B

Starting from Pythia-1B at step 1000, we freeze the embeddings and blocks 0–7 (half of the 1.01B parameters), which contain the eight selected heads (blocks 4–7). On target examples, we either

Table 27: Masking the rebuilt heads in suppressed-trained endpoints, in addition to the original set. Each cell gives rebuilt / layer-matched low-attention heads.

model	stream	added	held-out induction	target	key swap
410M	38	0	0.87 / 0.87	1.00 / 1.00	1.00 / 1.00
410M	38	2	0.19 / 0.87	0.85 / 1.00	0.74 / 1.00
410M	38	4	0.00 / 0.87	0.38 / 1.00	0.13 / 1.00
410M	38	8	0.00 / 0.87	0.05 / 1.00	0.01 / 1.00
410M	38	16	0.00 / 0.87	0.04 / 1.00	0.00 / 1.00
410M	39	0	0.84 / 0.84	1.00 / 1.00	0.99 / 0.99
410M	39	2	0.09 / 0.84	0.78 / 1.00	0.64 / 0.99
410M	39	4	0.00 / 0.84	0.26 / 1.00	0.05 / 0.99
410M	39	8	0.00 / 0.85	0.06 / 1.00	0.01 / 0.99
410M	39	16	0.00 / 0.84	0.04 / 1.00	0.01 / 0.99
1B	78	0	0.91 / 0.91	1.00 / 1.00	1.00 / 1.00
1B	78	2	0.63 / 0.85	0.99 / 1.00	0.99 / 1.00
1B	78	4	0.02 / 0.87	0.46 / 1.00	0.23 / 1.00
1B	78	8	0.00 / 0.88	0.01 / 1.00	0.00 / 1.00
1B	78	16	0.00 / 0.88	0.00 / 1.00	0.00 / 1.00
1B	79	0	0.94 / 0.94	1.00 / 1.00	1.00 / 1.00
1B	79	2	0.61 / 0.91	1.00 / 1.00	1.00 / 1.00
1B	79	4	0.00 / 0.91	0.25 / 1.00	0.10 / 1.00
1B	79	8	0.00 / 0.93	0.02 / 1.00	0.00 / 1.00
1B	79	16	0.00 / 0.93	0.02 / 1.00	0.00 / 1.00

Table 28: Latent induction heads in the pretrained checkpoints (step 1000). Attention is measured without suppression on the calibration probes; induction accuracy on a held-out probe, without and with the selected heads suppressed.

model	heads	suppressed	induction		other heads with attention		max other
	total		intact	suppressed	≥ 0.1	≥ 0.2	attention
160M	144	4	0.75	0.00	5	1	0.36
410M	384	8	0.64	0.00	8	5	0.58
1B	128	8	0.79	0.00	6	2	0.32
1.4B	384	7	0.81	0.01	15	8	0.67
2.8B	1024	4	0.75	0.00	15	6	0.56
410M, run 1	384	8	0.53	0.00	11	8	0.45
410M, run 2	384	4	0.75	0.00	7	3	0.47
410M, run 3	384	4	0.62	0.00	5	3	0.28

remove the heads’ contributions or replace them with the exact contributions of the frozen original (restoration). Ordinary induction and natural-text examples keep the original computation. Everything else follows the scaling protocol (learning rate 3×10^{-5} , streams 78/79).

Table 29 gives the outcomes. We extended the removal runs from 2,000 to 6,000 updates after partial learning appeared at 2,000; the controls ran for 2,000 updates. Restoration reproduces the trajectories of the intact runs exactly.

The removal-trained models pass the key tests only when evaluated with the heads still removed. Their ordinary induction under removal stays at or below 2/256 on our probe. This does not show that the workaround uses no retrieval: it may use computations that our ordinary-induction probe does not measure. We therefore describe it as a key-dependent workaround built in the trainable upper layers, not as learning without the prerequisite.

O REMOVING ATTENTION TO THE KEY

This intervention is stronger in kind and weaker in interpretation than head suppression, so we report it separately. On mapped-copy training examples only, it blocks, in every layer and head, attention from the source marker and every later position to the four tokens of the earlier key occurrence.

Table 29: Pythia-1B with the lower half frozen. *Removal on* evaluates the model with the heads still removed; *native* evaluates it with the original heads.

condition (streams 78/79)	first learned	horizon	end target	end key swap
restoration, native	300 / 300	2,000	0.984 / 0.980	0.992 / 0.973
removal, removal on	2,750 / 2,800	6,000	0.980 / 0.980	0.945 / 0.960
removal, native	never	6,000	0.023 / 0.039	0.000 / 0.000

Attention to the source marker itself and to the second key occurrence remains. The control applies the same cut to the nearest irrelevant four-token block. Ordinary induction and natural-text examples are unaffected, and evaluation is unmasked.

With learning rate 3×10^{-5} over 2,000 updates, the cut blocks key-dependent learning at 160M, 410M, 1B and 1.4B on both streams. It never meets the key-dependence criterion at any of the 41 evaluations, while every intact and control run learns by update 150–350. At 410M this holds even though the models learn by update 900 under the fixed eight-head suppression on the same streams.

We do not treat this as evidence about prerequisite circuits, for two reasons. First, the cut removes information the target needs, namely the earlier key, rather than a learned computation, so blocking is close to guaranteed unless some circuit reads the key indirectly. Second, the cut sits at the answer’s key, so the pattern of blocked positions itself reveals where the source is. A model given the mask geometry and the mapping could solve the task without reading the key, and it would fail the key-deletion test.

The models trained with the cut do improve their target likelihood without learning the task. Most of this improvement comes from avoiding the source marker’s unmapped identity (4.6–5.9 nats), with a smaller preference for the correct marker (1.2–1.6 nats). A restoration experiment at 410M finds that most of this benefit carries over to a new mapping, which suggests that the models trained with the cut learn mostly mapping-independent marker statistics.

P ADDITIONAL SEEDS OF THE TWO-LAYER MODEL

To add seeds to the suppression and rescue experiments, we train five new models (seeds 3541–3545) from scratch with the original warm-up, formation rule, head selection rule and control-head rule. We then run the unsuppressed, control-suppressed, suppressed, correct-supply, no-supply and wrong-supply conditions on the same 1,000 target batches, with the original exact-restoration checks (frozen weights unchanged, restored computation bit-identical at every evaluation). All five models form the circuit, and in each the largest-effect head is a first-layer head.

Table 30 gives every endpoint. Suppression blocks learning in all five, and supplying the intact activation restores it to the unsuppressed value. The unsuppressed copy gain varies more across these models (1.25–2.51 nats) than across the original three. In one model (3545) the control-head set removes more short-range accuracy than the original floor would allow (0.49 against 0.60); the original code records but does not select on this check, and we report the model unchanged. In none of the new models does replay with both heads enabled reach 1 nat within 1,000 updates after suppressed exposure (maximum 0.30), which is consistent with the mixed replay outcomes in Table 6.

These runs used a different GPU and library version from the originals. Retraining seed 531 from scratch reproduces its inputs bit-for-bit and the qualitative pattern, but not its exact published values, because floating-point differences change the model in which the circuit forms.

Q FURTHER EXPERIMENTS

Finer interventions. Head-level suppression removes everything a head computes. Suppressing induction at the level of attention edges between specific positions, QK features (Kamath et al., 2025) or learned subspaces, with circuits found by attribution methods such as EAP-IG or edge pruning (Hanna et al., 2024; Bhaskar et al., 2024) and validated by activation patching (Zhang and Nanda, 2024), would show whether the model also routes around a narrower cut, and would reduce

Table 30: Five additional seeds of the two-layer model: copy gain (nats) / distance-64 accuracy after 1,000 target updates. *Replay* is copy gain after 1,000 further updates of exact replay with both heads enabled, following suppressed exposure.

seed	formed at	head	unsuppressed	control	suppressed	correct supply	no supply	wrong supply	replay
3541	44,500	1.1	2.23 / 0.46	2.19 / 0.42	-0.01 / 0.00	2.23 / 0.46	-0.01 / 0.00	-0.01 / 0.00	-0.01
3542	78,500	1.4	1.25 / 0.18	1.18 / 0.19	0.01 / 0.01	1.25 / 0.18	0.01 / 0.01	0.01 / 0.01	0.01
3543	43,500	1.7	2.03 / 0.23	2.01 / 0.22	0.00 / 0.03	2.03 / 0.23	0.00 / 0.03	0.00 / 0.03	0.00
3544	45,500	1.7	2.00 / 0.38	1.97 / 0.37	0.02 / 0.00	2.00 / 0.38	0.02 / 0.00	0.02 / 0.00	0.03
3545	102,500	1.7	2.51 / 0.36	2.49 / 0.32	0.02 / 0.01	2.52 / 0.36	0.02 / 0.01	0.02 / 0.01	0.30

collateral effects. The key-triggered adaptive procedure in Section 5 is a coarse version of such computation-level suppression.

Predicting when data will teach. The rebuilt heads are identifiable before training by their induction attention. A readiness score built from such latent capacity (for instance, the attention and copying scores of unused heads) could predict whether and how quickly a model will learn from a given data stream, and could guide data scheduling. Testing such a predictor prospectively across PolyPythias runs and checkpoints is a direct next step.

Controlled scaling. Pythia sizes differ in depth, width, head width and the maturity of a checkpoint at a given step. Training small families that differ only in heads per layer or width, forming the same prerequisite and applying the same computation-level suppression would test whether the delay falls with the number of candidate backup heads, as our backup-head account predicts. The escape-dimension view of overparameterization (a preprint; Martinelli et al., 2026) would suggest a similar dependence on width through a different mechanism. Such families would also allow direct measurement of the loss landscape during the delay (for example, the plateau and the curvature along the parameters of the rebuilt heads).

Suppressing the previous-token step in pretrained models. In the two-layer model the previous-token step cannot be rebuilt, while the induction step can. Our interventions in pretrained models suppress the induction heads. Suppressing previous-token heads on target examples in Pythia would test whether the previous-token step is also harder to rebuild at scale.

Other prerequisite families. Candidate pairs include entity binding as a prerequisite for multi-hop retrieval, successor or carry circuits for multi-digit arithmetic, and syntactic heads for agreement. Each admits the same fixed-data design: training-only suppression, activation rescue and exact replay.

Natural pretraining. The Pythia suite’s public data order makes it possible to intervene on circuits during pretraining itself and ask which later capabilities are delayed, rather than using synthetic fine-tuning targets.

Robust capability removal. In larger models, suppressing a prerequisite delays but does not prevent learning. Whether adaptive, computation-level suppression can keep a capability from being learned, and at what cost to other capabilities, bears directly on tamper-resistant unlearning (Tamirisa et al., 2025; Che et al., 2025).